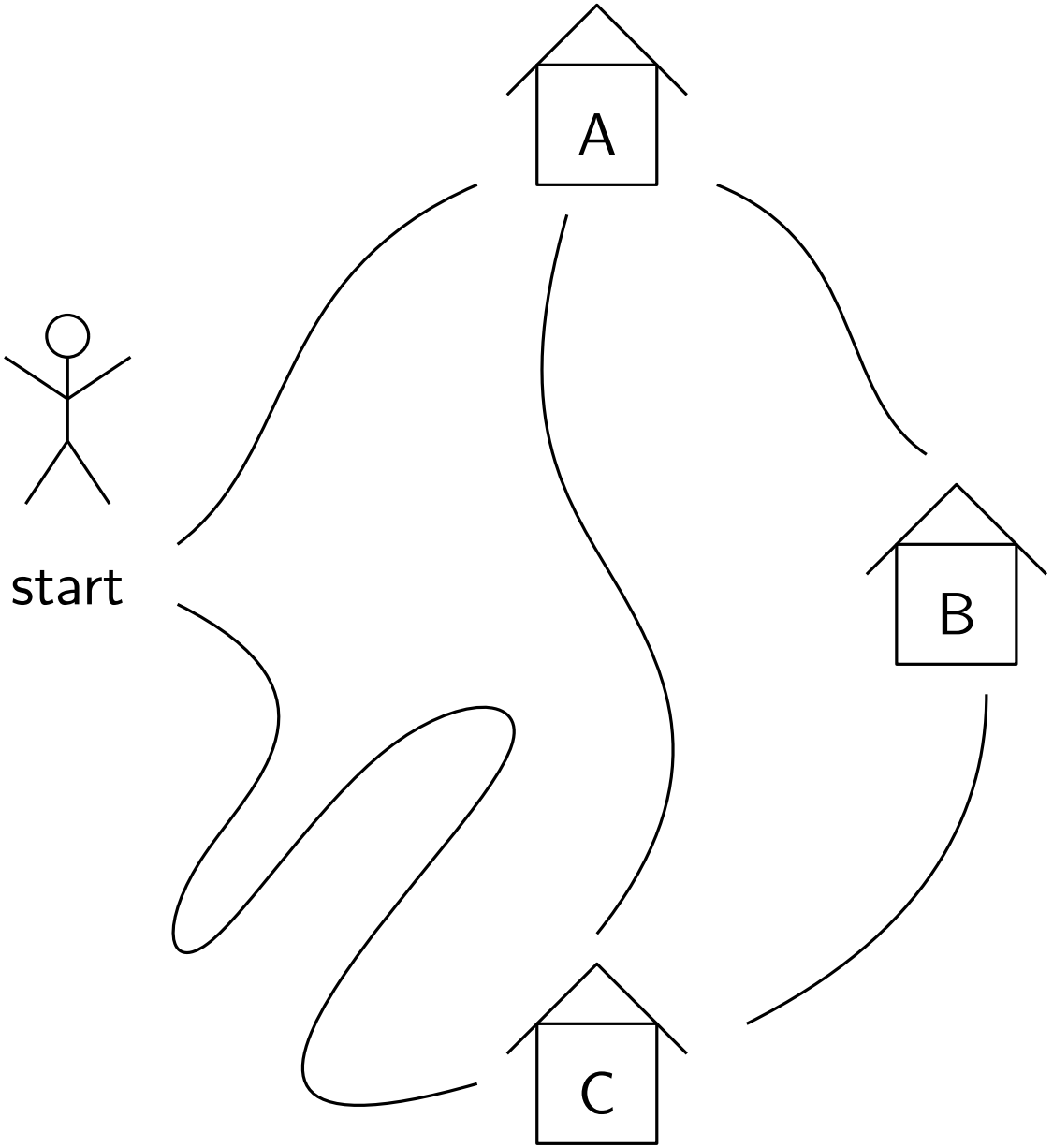
Ein Einblick in die Kominatorische Optimierung und in die Kominatorische Geometrie

Manfred Scheucher

Optimaler Reiseweg durch die 15 größten Städte Deutschlands.
https://de.wikipedia.org/wiki/Problem_des_Handlungsreisenden

ZWISCHENFRAGEN ERWÜNSCHT!

Das Rundreiseproblem



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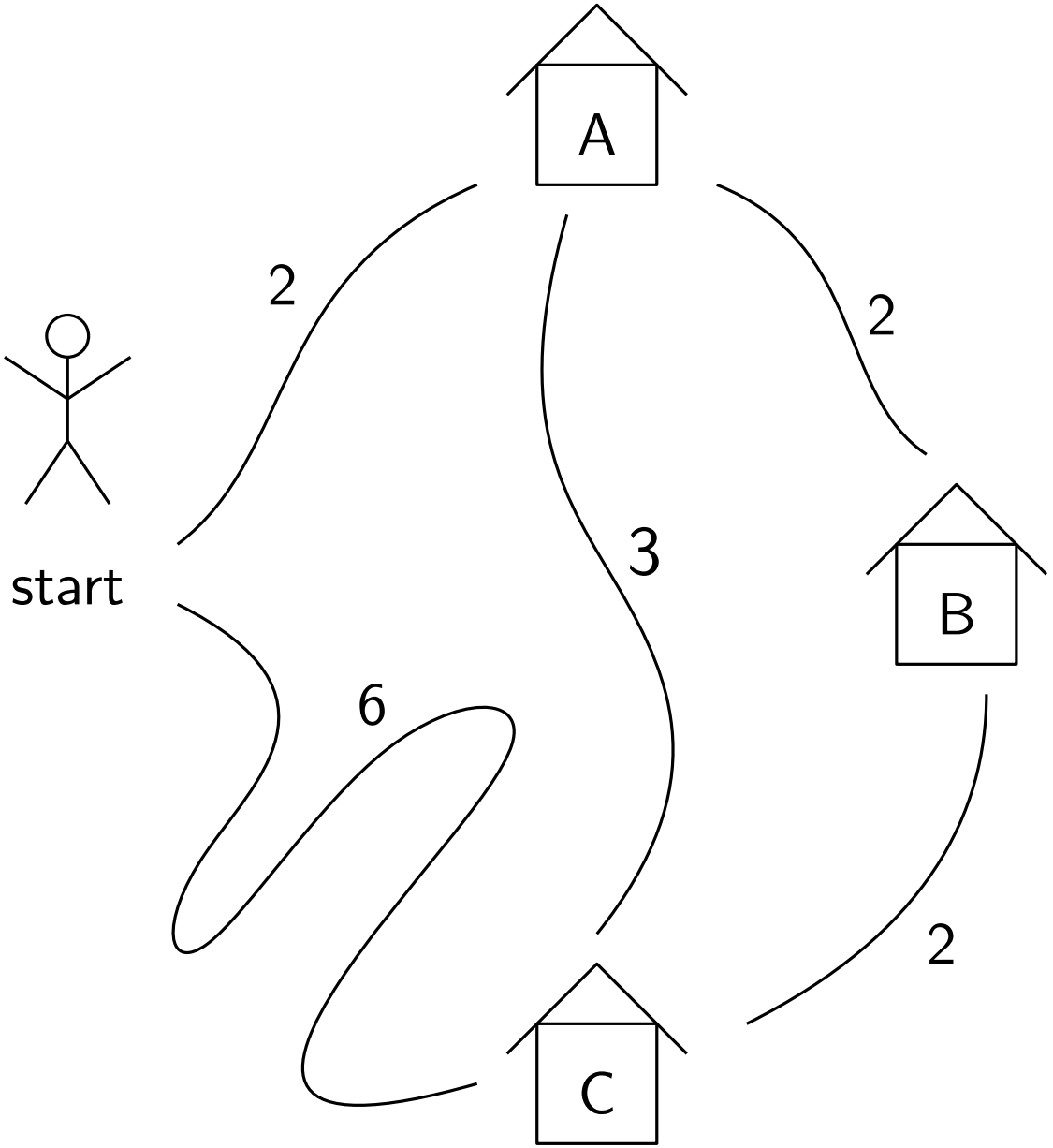

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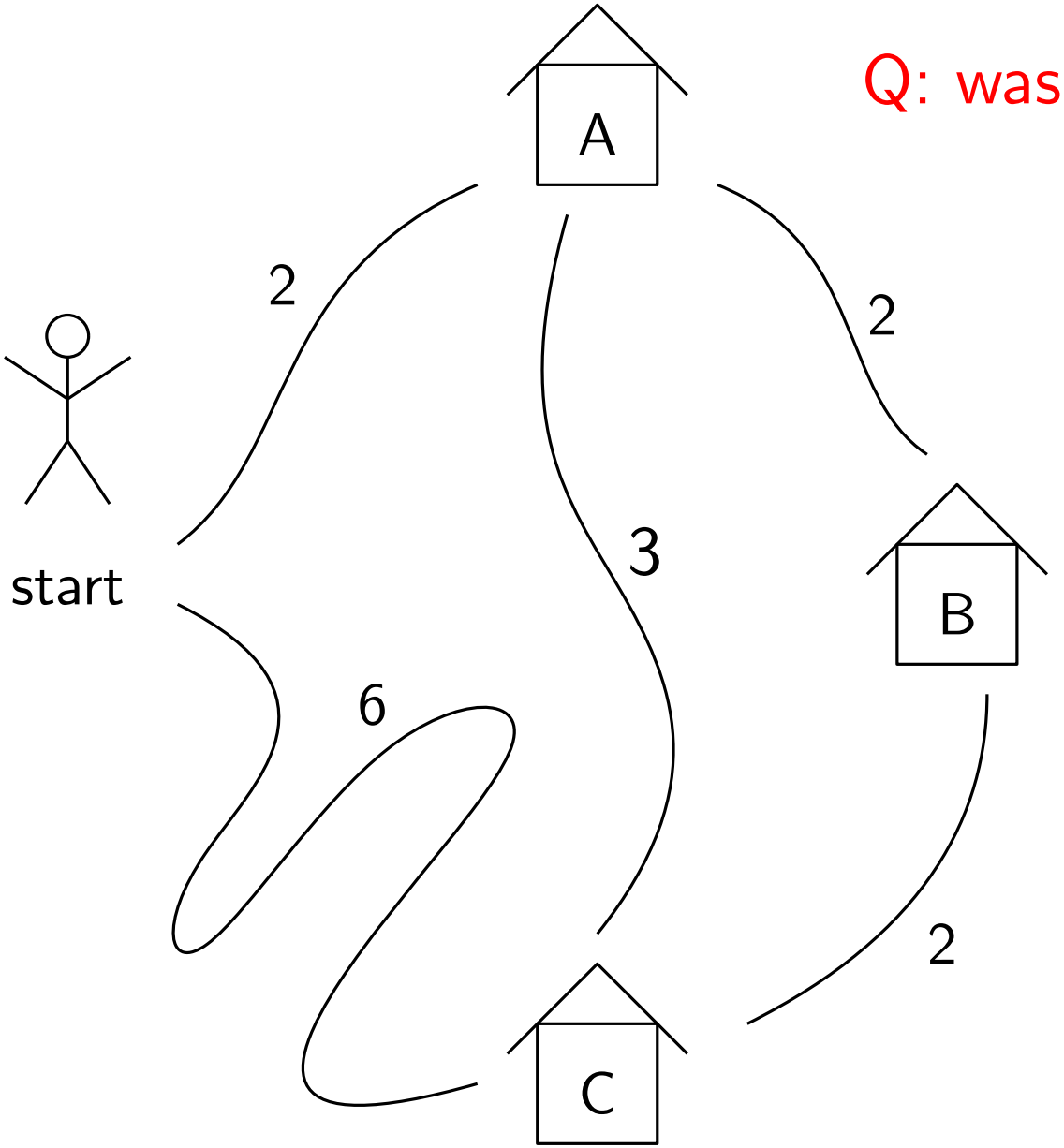
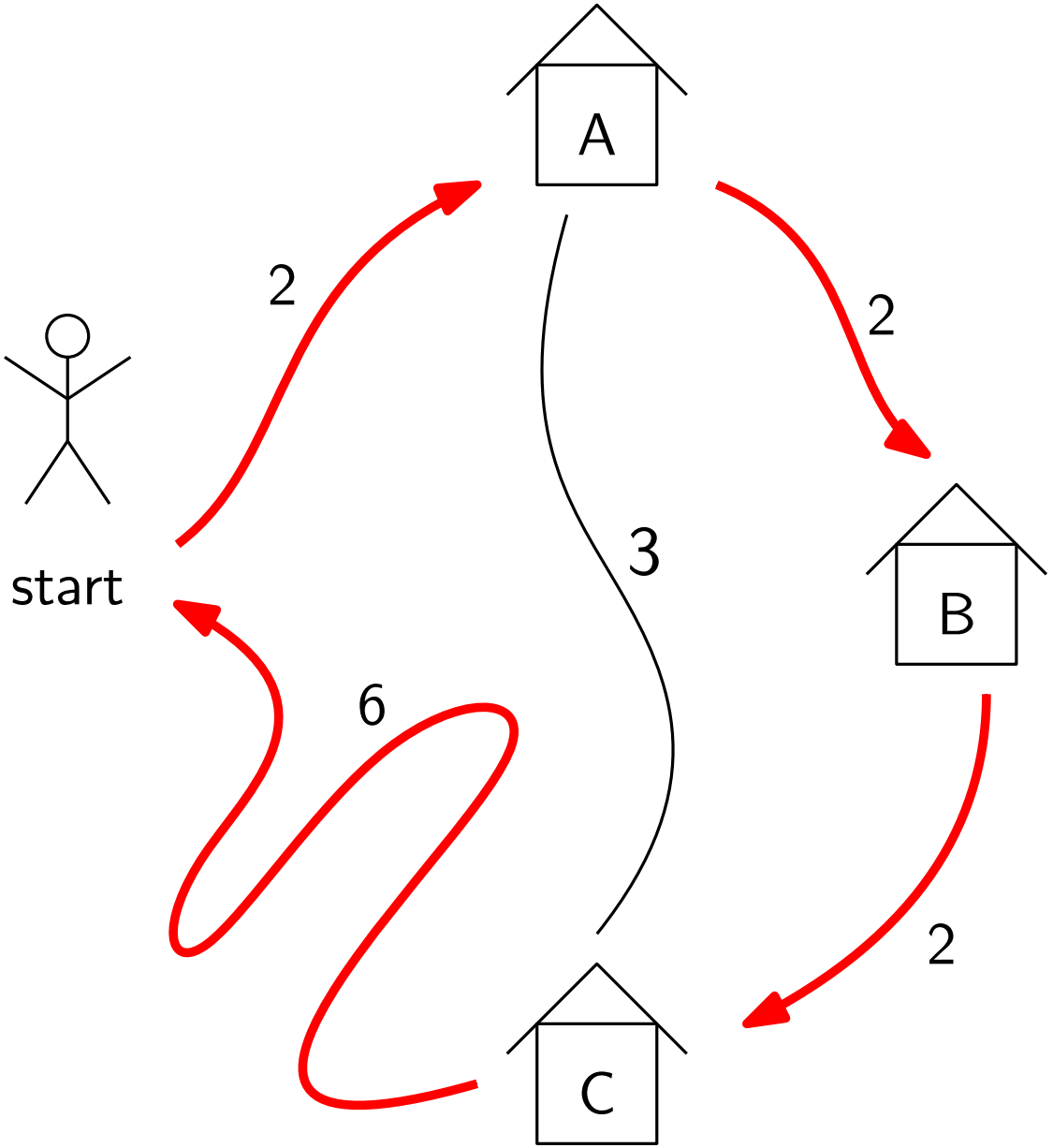


Q: was ist die kürzeste Route?

Das Rundreiseproblem

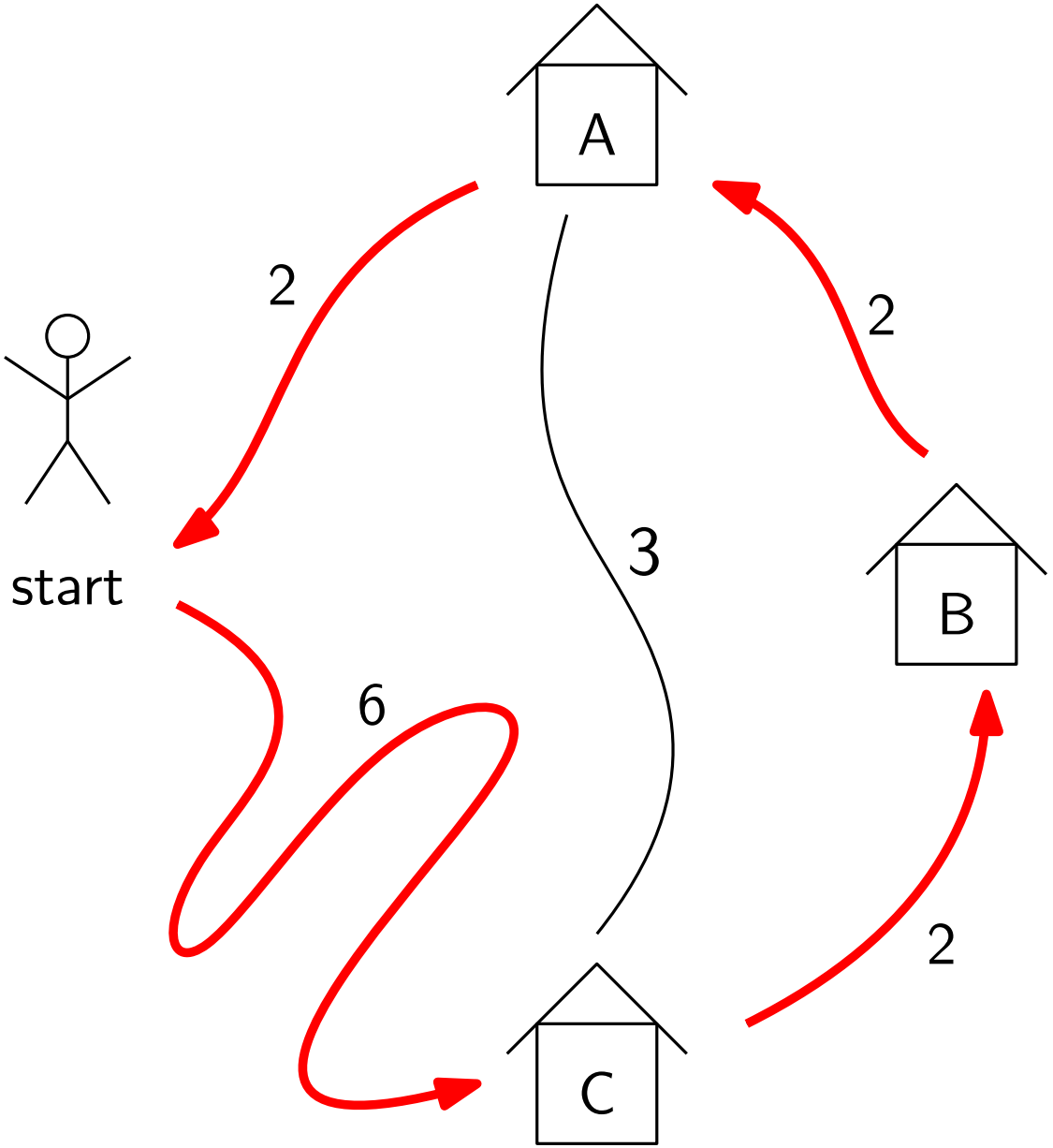


Reiserouten:

sABCs

2+2+2+6=12

Das Rundreiseproblem



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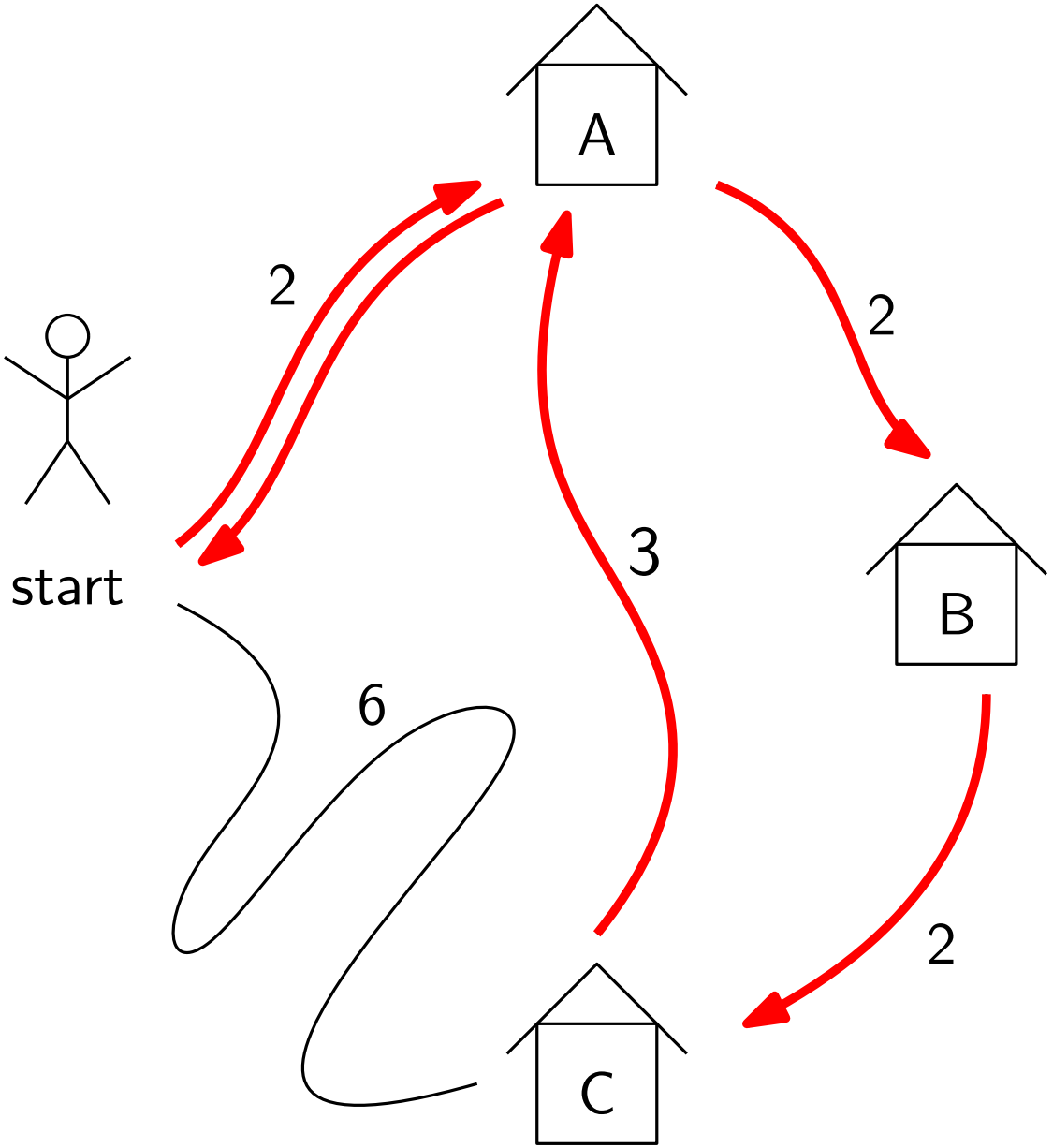

Reiserouten:

sABCs
2+2+2+6=12

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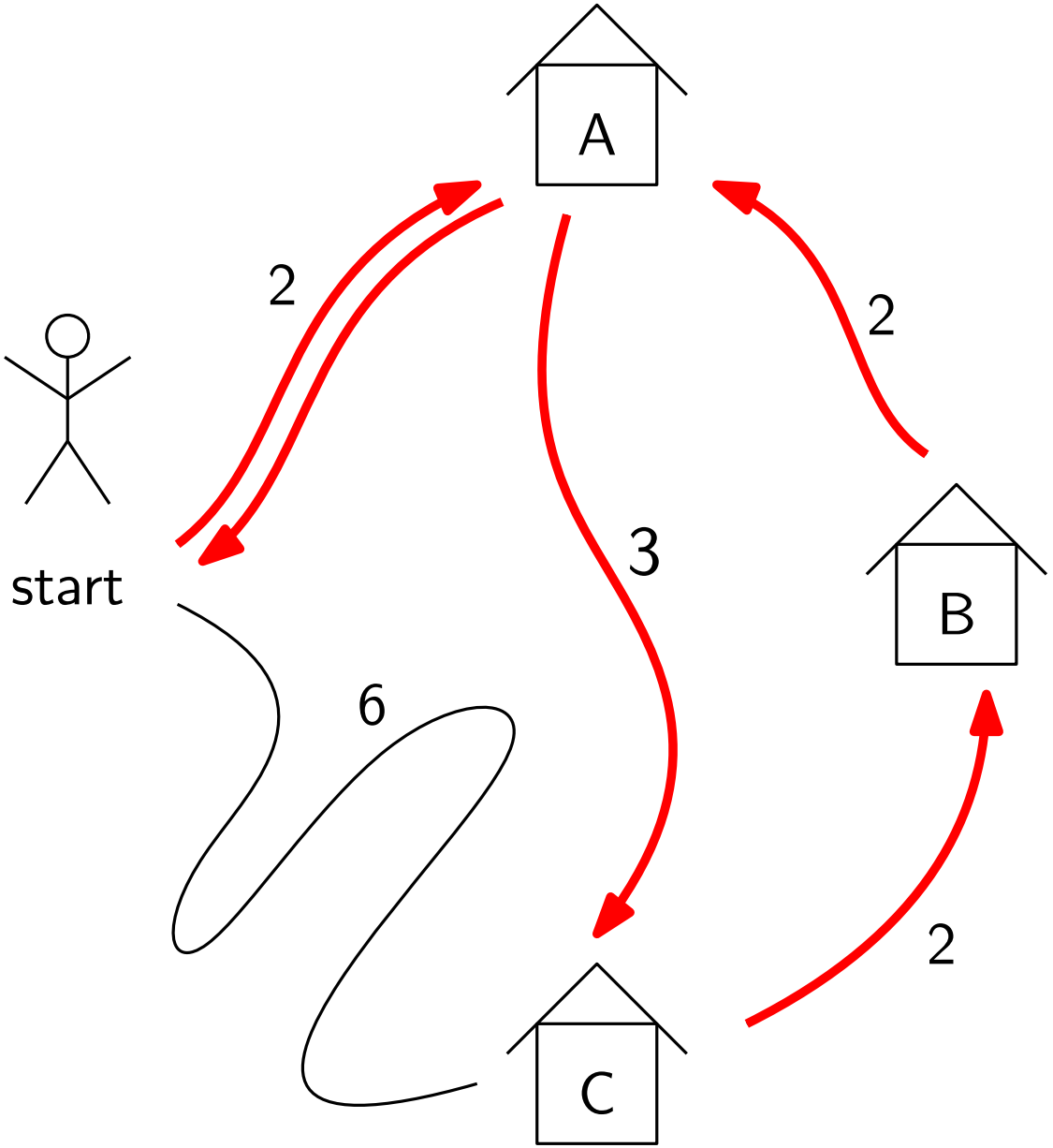
Städte mehrfach:

sABC(A)s
2+2+2+3+2=11

Das Rundreiseproblem



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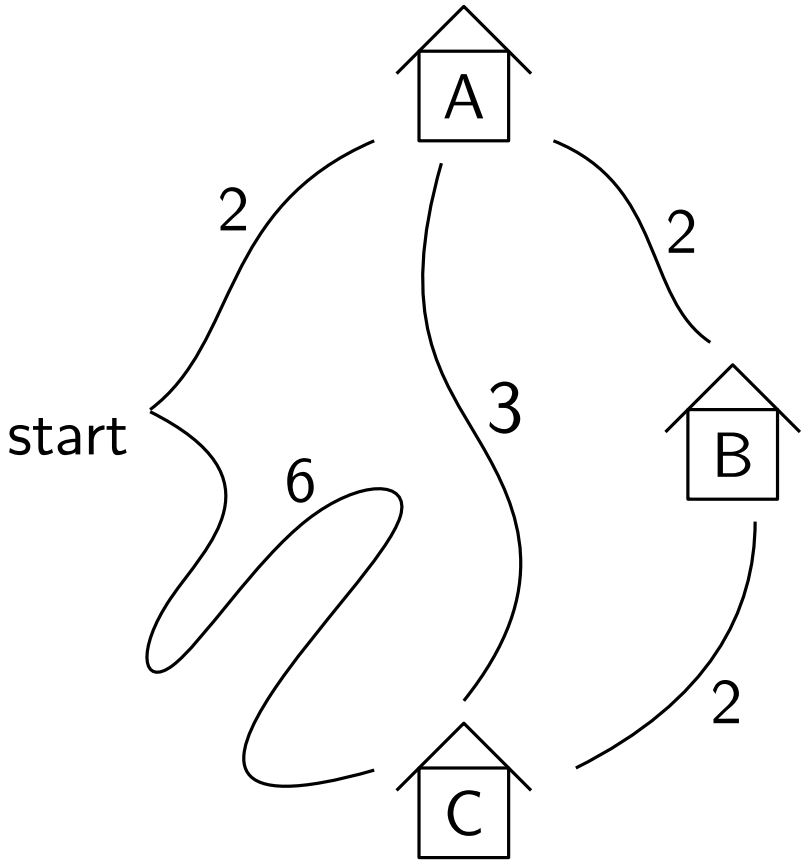
Das Rundreiseproblem

- Modellierung als vollständiger Graph
  (d.h. alle Destinationen sind verbunden)



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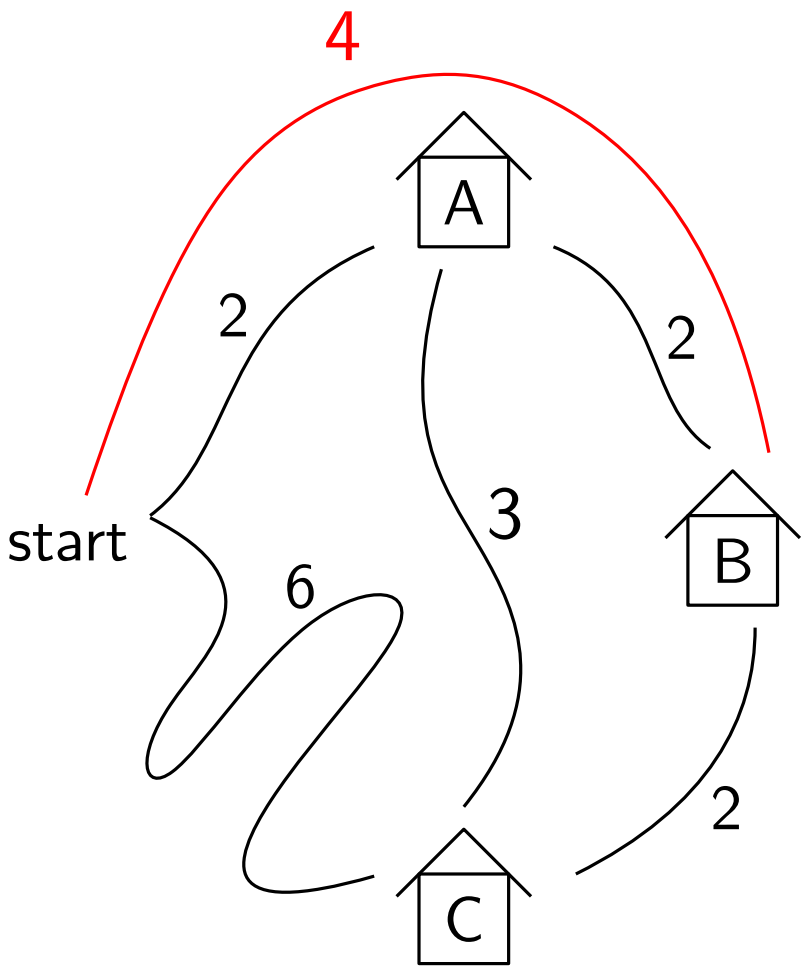

- je 2 Destinationen verbinden

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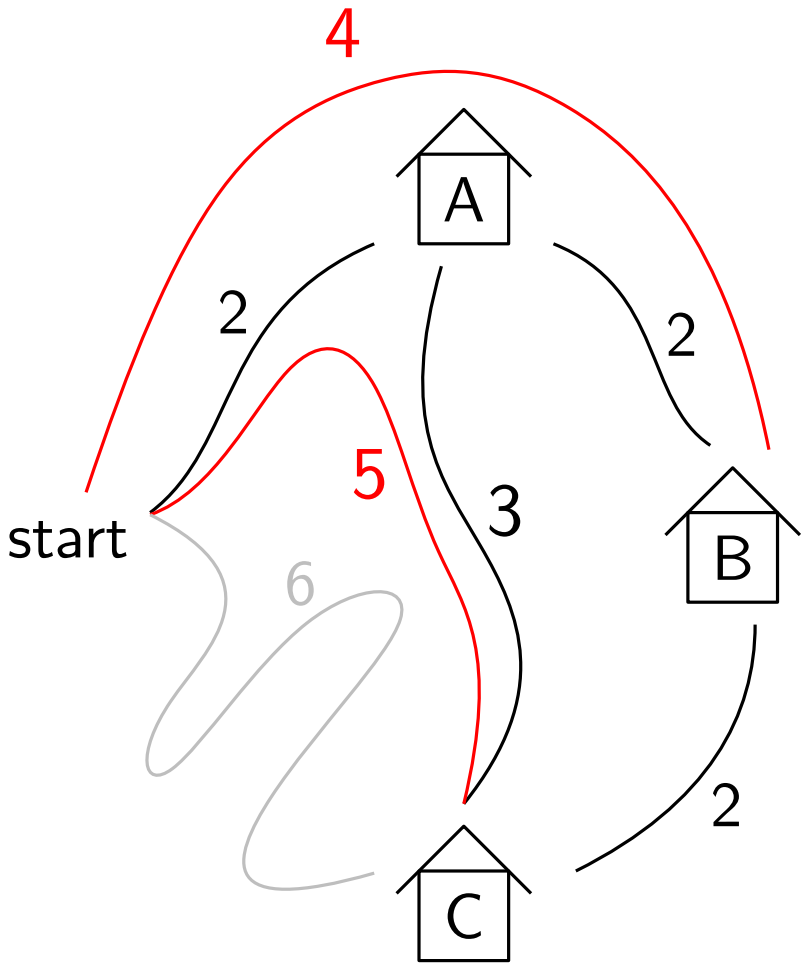

- je 2 Destinationen verbinden
- Dreiecksungleichung:
  $d(X, Y) \leq d(X, Z) + d(Z, Y)$
  für jeden Zwischenstopp Z

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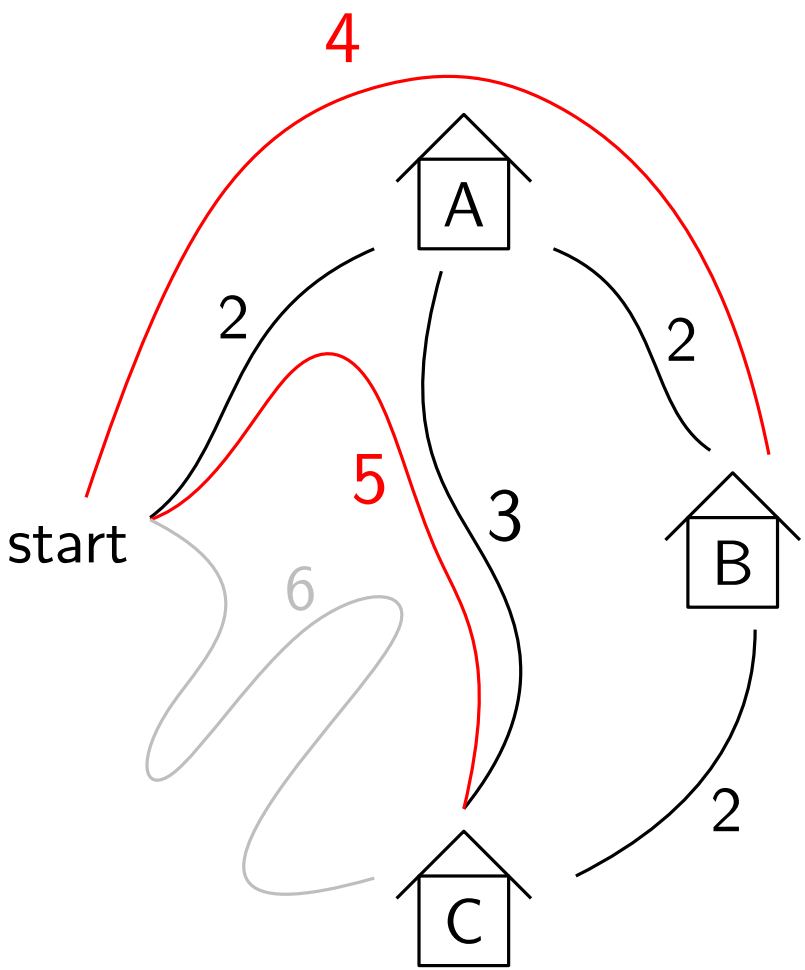

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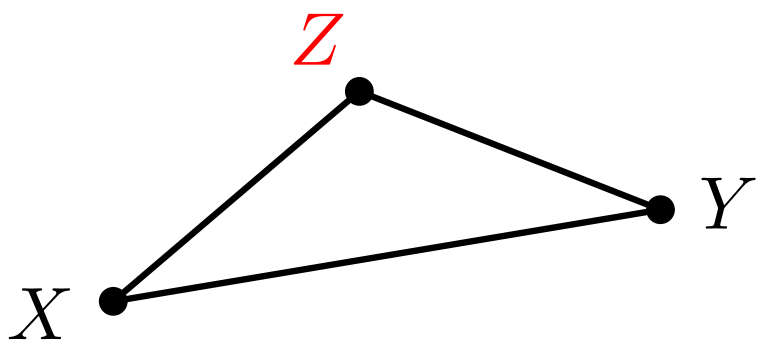

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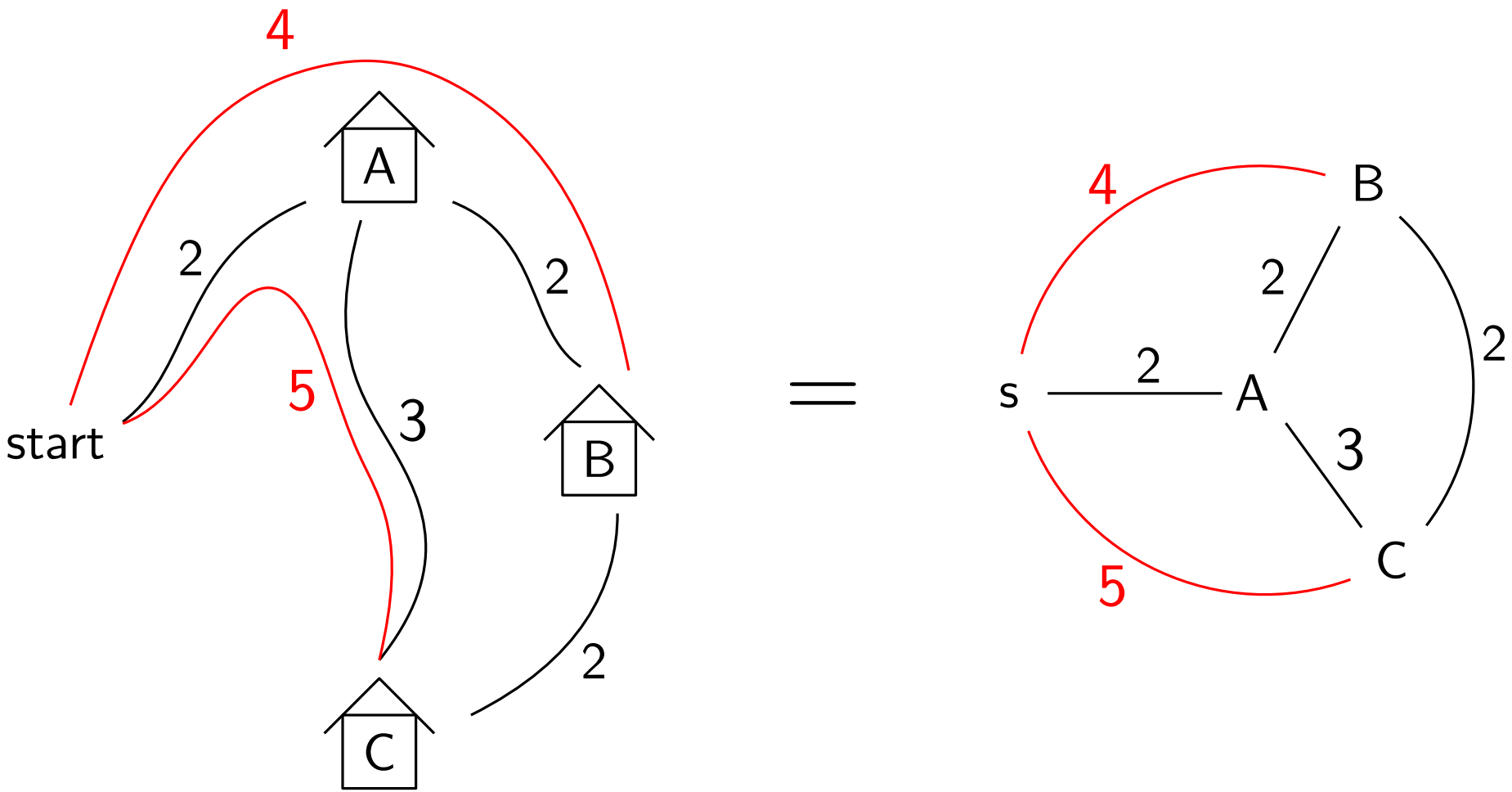
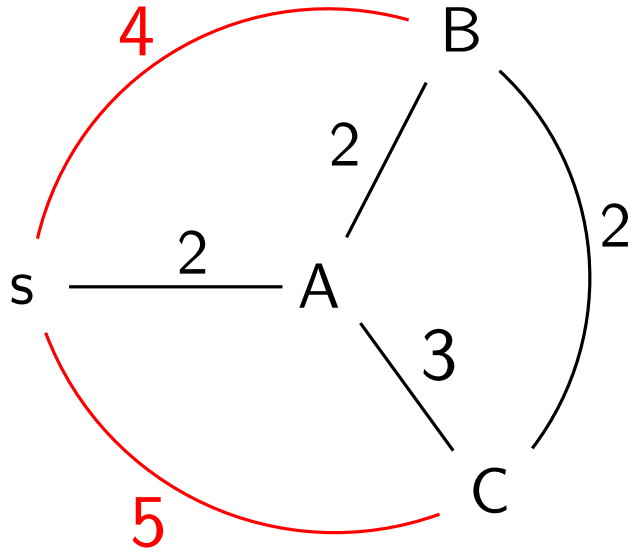


Das Rundreiseproblem

- Modellierung als vollständiger Graph
  (d.h. alle Destinationen sind verbunden)

- $3! = 3 \cdot 2 \cdot 1 = 6$ Kandidaten für optimale Runden
  (Hamiltonkreise)

sABCs
sACBs
sBACs
sBCAs
sCABs
sCBAs



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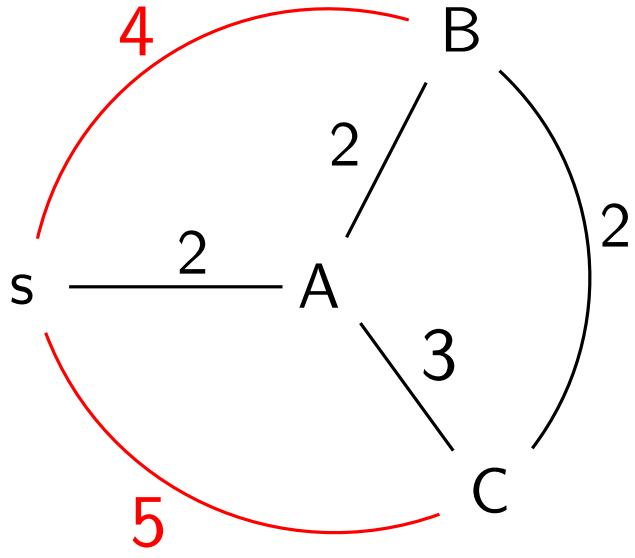
sABCs	11
sACBs	11
sBACs	14
sBCAs	11
sCABs	14
sCBAs	11



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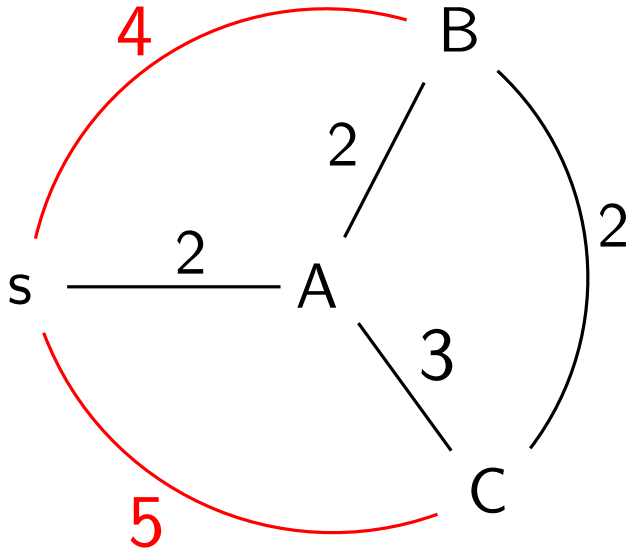


- NP-schweres Problem (1 der 7 Milleniumprobleme)

Das Rundreiseproblem

- Modellierung als vollständiger Graph

TSP : sage-ipython — Konsole

File Edit View Bookmarks Plugins Settings Help

New Tab Split View Copy Paste Find

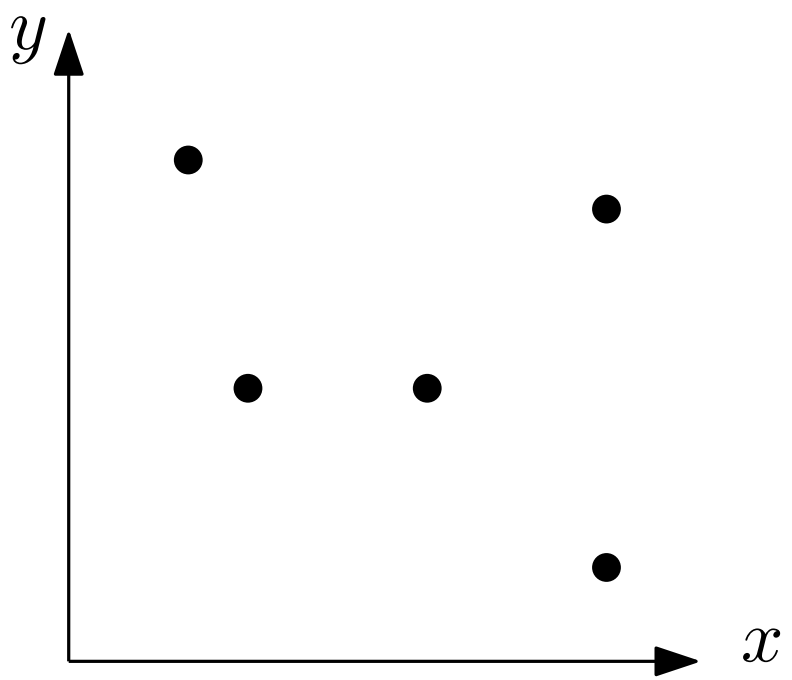
```
sage: G = Graph([
....:     ['S','A',2],
....:     ['S','B',4],
....:     ['S','C',5],
....:     ['A','B',2],
....:     ['A','C',3],
....:     ['B','C',2],
....:     ])
sage: tsp = G.traveling_salesman_problem(
....:     use_edge_labels=True)
sage: print(tsp.edges())
[('A', 'C', 3), ('A', 'S', 2), ('B', 'C', 2), ('B', 'S', 4)]
sage: print(sum(tsp.edge_labels()))
11
sage:
```

2

- NP-schweres Problem (1 der 7 Milleniumprobleme)

Euklidisches Rundreiseproblem

- Destinationen als Punkte, euklidische Abstände
- $d(A, B) = \|B - A\|_2 = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$



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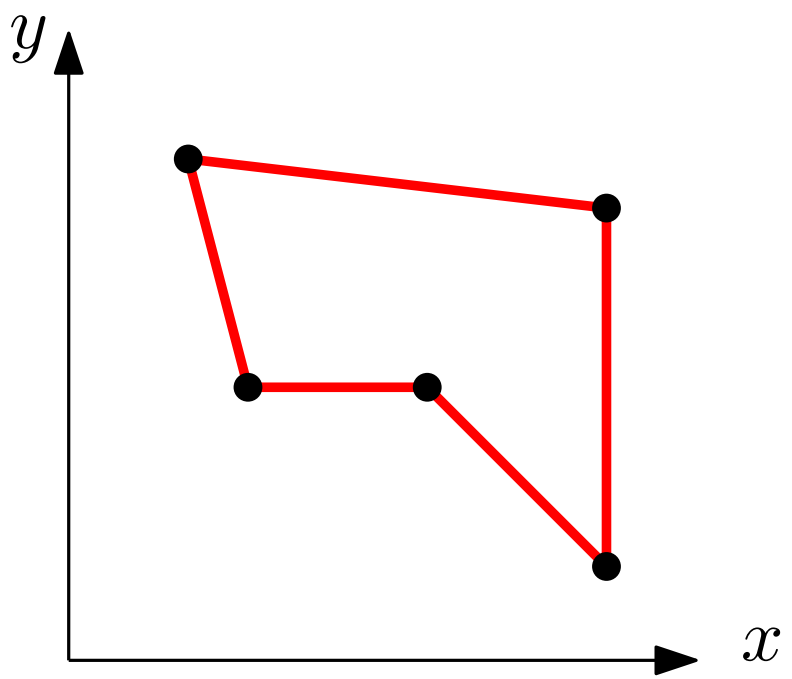

Euklidisches Rundreiseproblem

- Destinationen als Punkte, euklidische Abstände
- $d(A, B) = \|B - A\|_2 = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$
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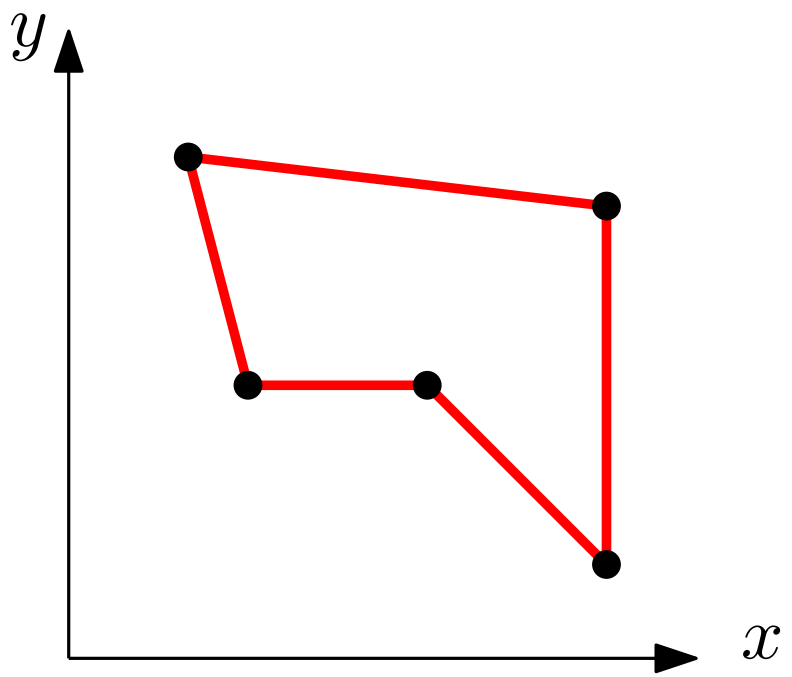


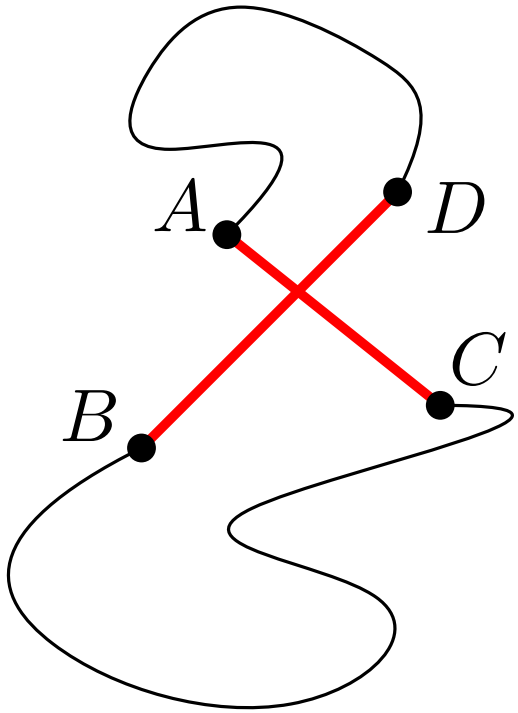
Euklidisches Rundreiseproblem

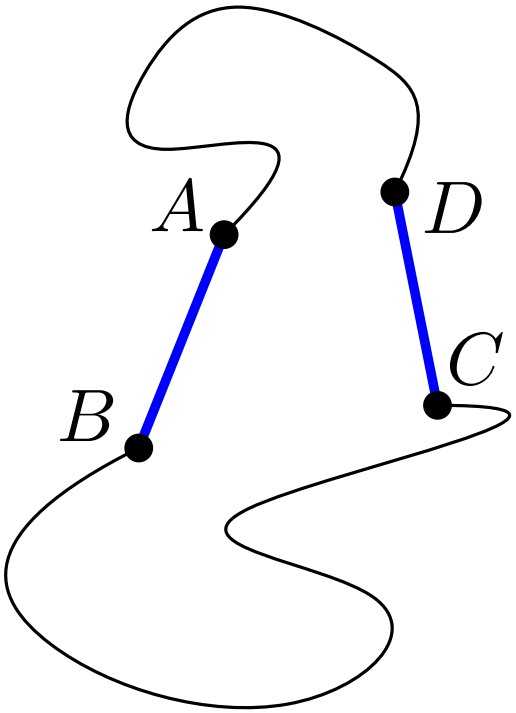
Beispiel: Kürzester Rundreiseweg durch die 15 größten Städte Deutschlands.

Quelle: `https://de.wikipedia.org/wiki/Problem_des_Handlungsreisenden`

Euklidisches Rundreiseproblem

- Dreiecksungleichung $\Rightarrow$ optimale Routen *kreuzungsfrei*





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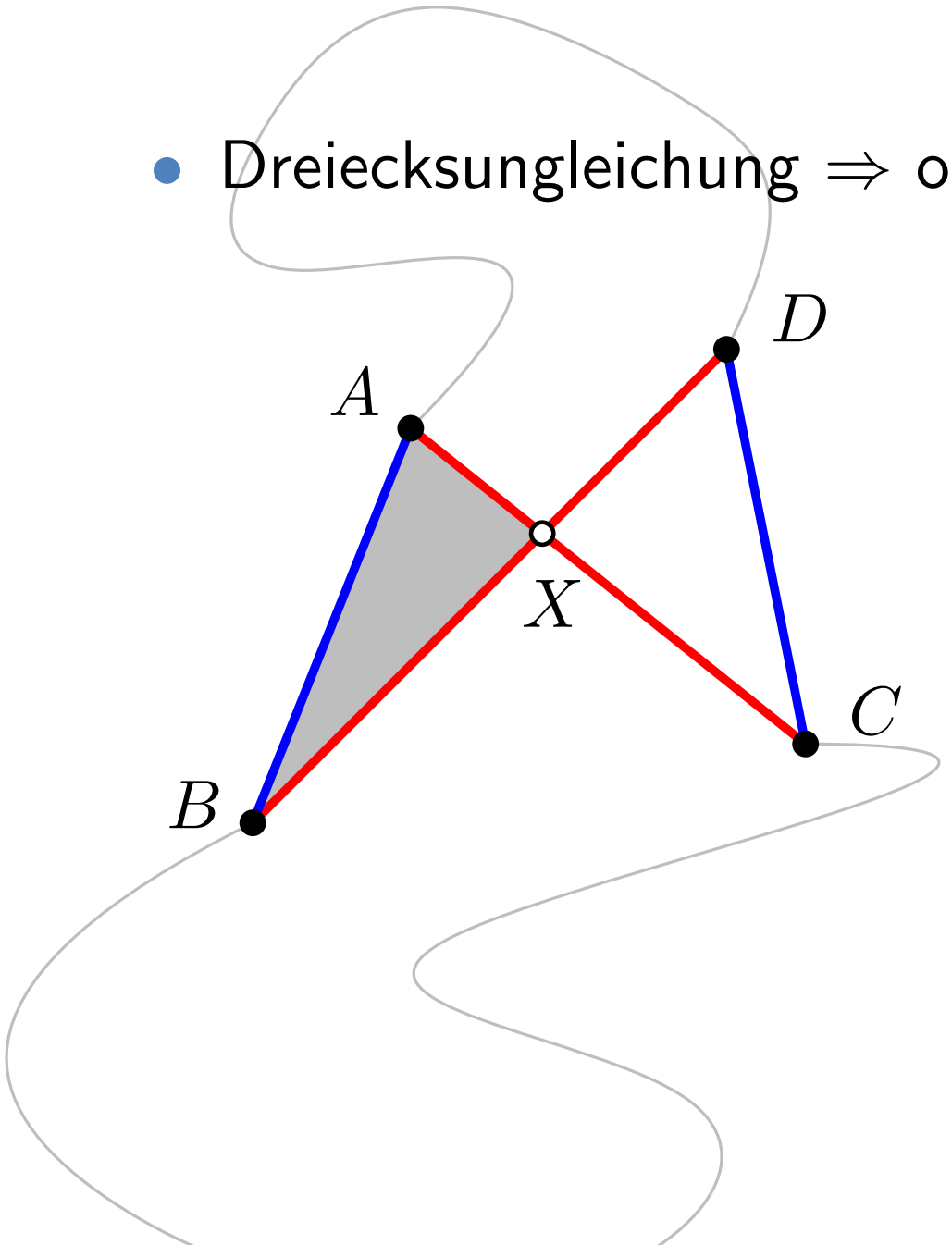

$$d(A,B) \leq d(A,X) + d(X,B)$$

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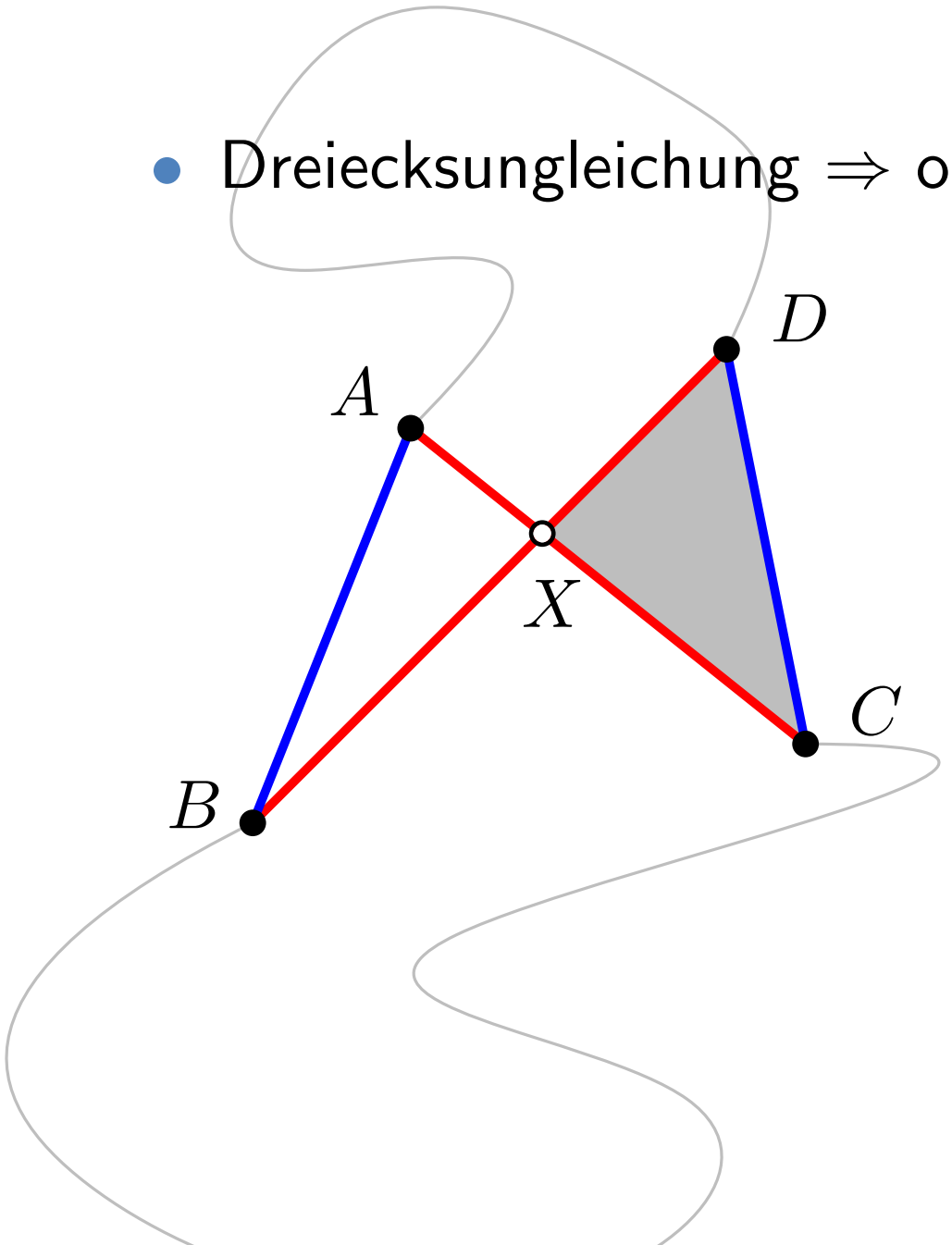
- Dreiecksungleichung $\Rightarrow$ optimale Routen *kreuzungsfrei*



$$d(A,B) \leq d(A,X) + d(X,B)$$

$$d(C,D) \leq d(C,X) + d(X,D)$$

Euklidisches Rundreiseproblem

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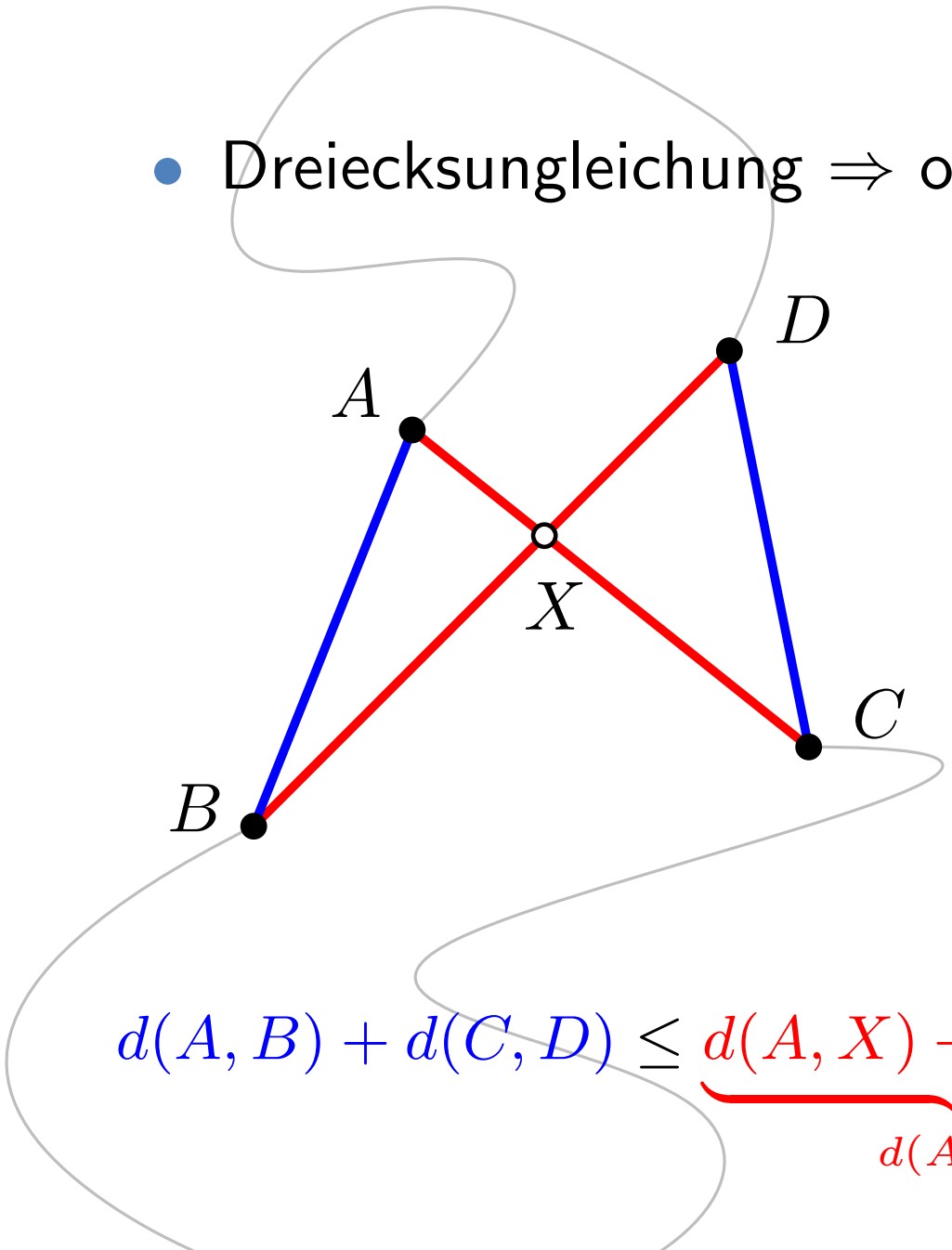

$$d(A,B) \leq d(A,X) + d(X,B)$$

$$d(C,D) \leq d(C,X) + d(X,D)$$

$$d(A,B) + d(C,D) \leq \underbrace{d(A,X) + d(X,C)}_{d(A,C)} + \underbrace{d(B,X) + d(X,D)}_{d(B,D)}$$

Euklidisches Rundreiseproblem

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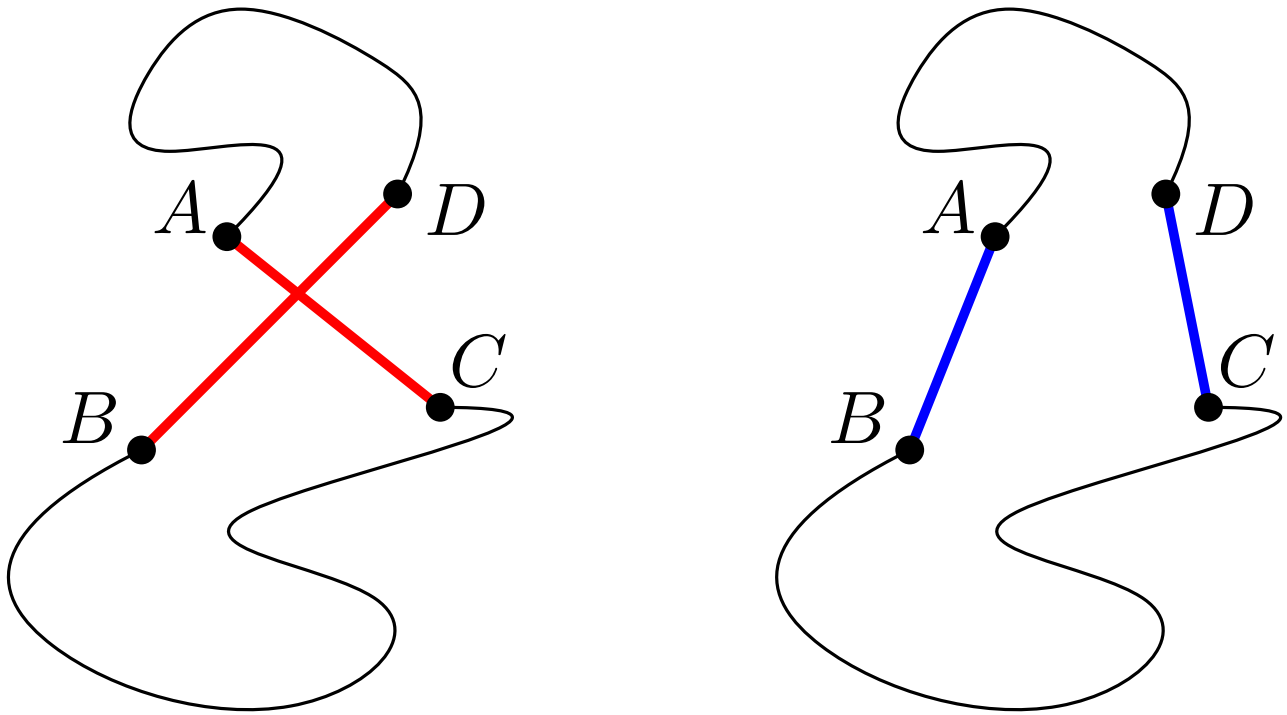

$$d(A,B) + d(C,D) \leq \underbrace{d(A,X) + d(X,C)}_{d(A,C)} + \underbrace{d(B,X) + d(X,D)}_{d(B,D)}$$

Euklidisches Rundreiseproblem

- Dreiecksungleichung $\Rightarrow$ optimale Routen *kreuzungsfrei*
- Es gibt $(n-1)! = (n-1) \cdot (n-2) \cdot \ldots \cdot 2 \cdot 1$ Touren.
  Wie viele sind kreuzungsfrei (Kandidaten für optimal)?

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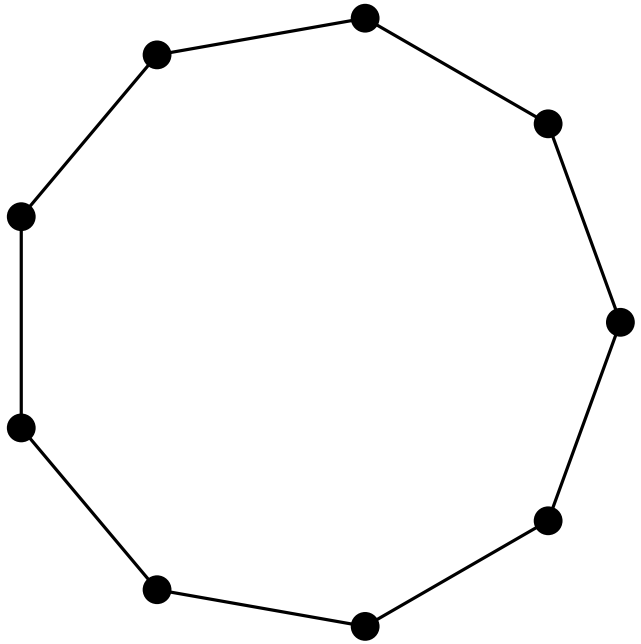

- eindeutige Tour wenn Punkte in *konvexer Lage*

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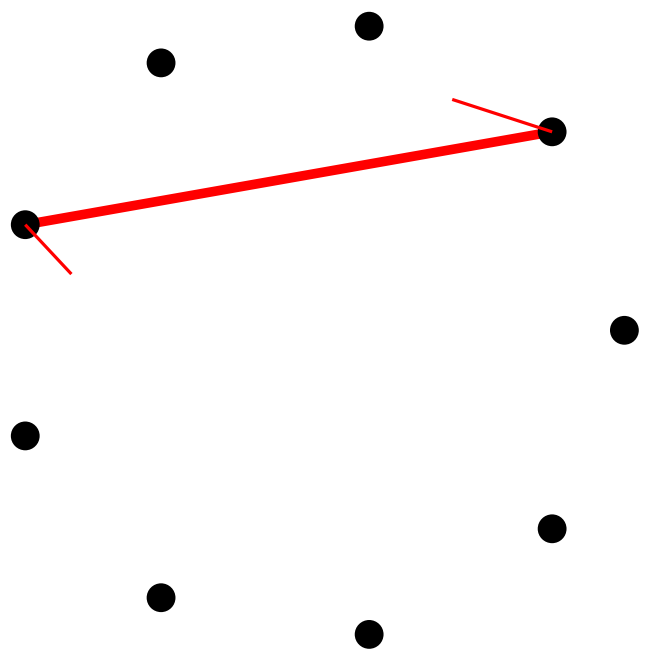


- eindeutige Tour wenn Punkte in *konvexer Lage*

  (keine Diagonale in kreuzungfreier Tour)

Euklidisches Rundreiseproblem

- Dreiecksungleichung $\Rightarrow$ optimale Routen *kreuzungsfrei*
- Es gibt $(n-1)! = (n-1) \cdot (n-2) \cdot \ldots \cdot 2 \cdot 1$ Touren.
  Wie viele sind kreuzungsfrei (Kandidaten für optimal)?
- $\#$ von Kombinatorik der Punktmenge abhängig

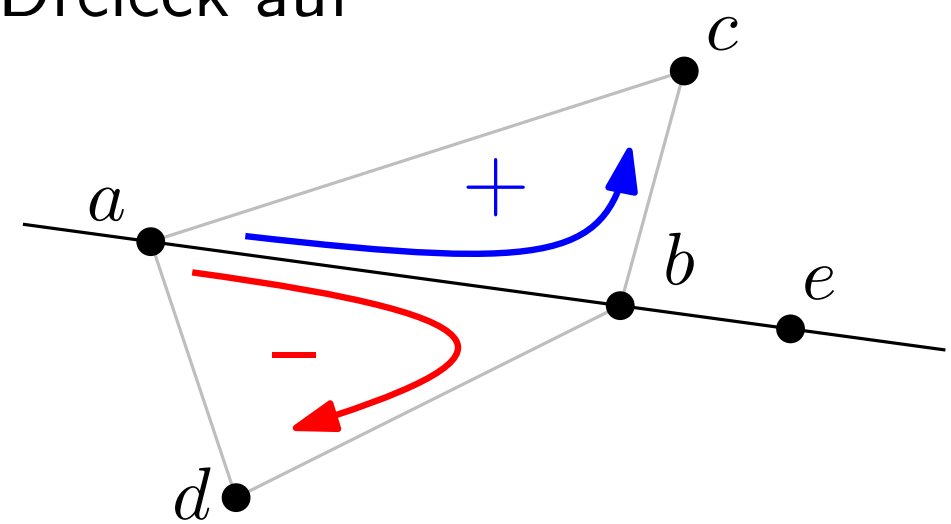
Punktkonfigurationen

Seien $p_1, \ldots, p_n$ Punkte mit Koordinaten $p_i = (x_i, y_i)$

Je 3 Punkte spannen orientiertes Dreieck auf

$\chi : [n]^3 \to \{+, 0, -\}$

"positive, collinear, negative"



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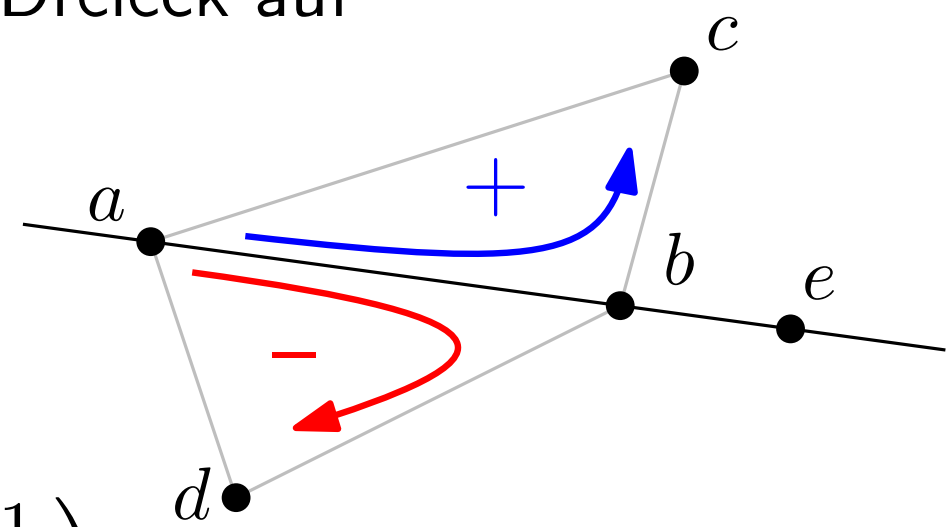

Formell:

$$\chi(a,b,c) := \operatorname{sgn}\det\begin{pmatrix} 1 & 1 & 1 \\ x_a & x_b & x_c \\ y_a & y_b & y_c \end{pmatrix}$$

$$= (x_b - x_a)\cdot(y_c - y_a) - (x_c - x_a)\cdot(y_b - y_a)$$

Punktkonfigurationen

Seien $p_1, \ldots, p_n$ Punkte mit Koordinaten $p_i = (x_i, y_i)$

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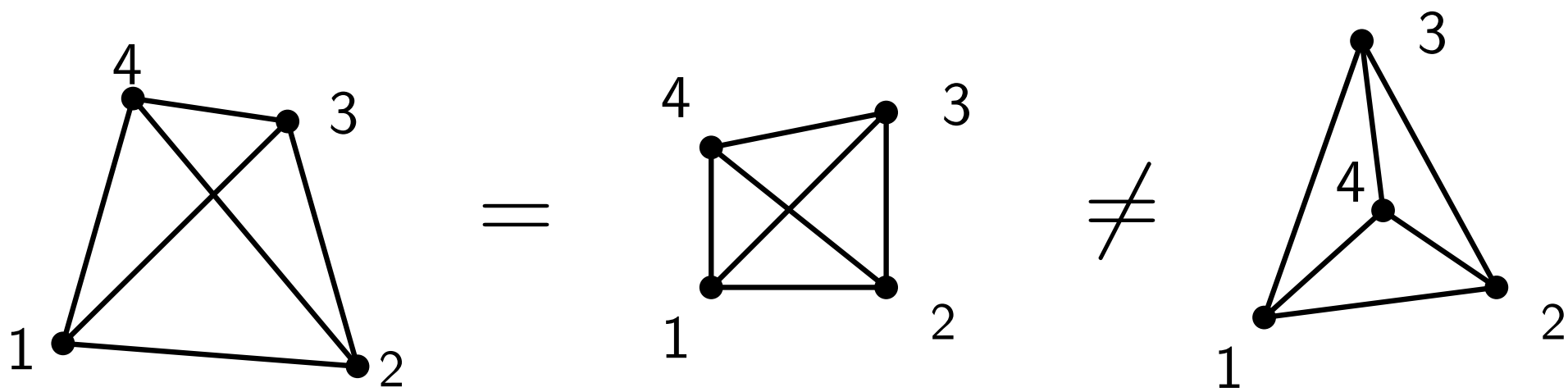
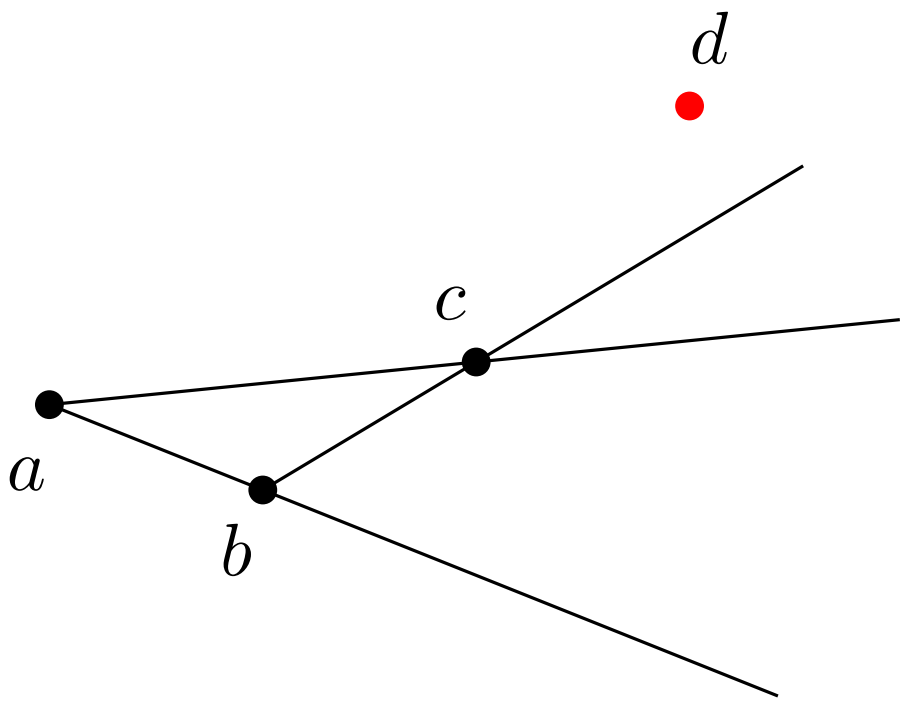


Zwei Punktmengen P und Q sind vom selben Typ
wenn $\chi_P = \chi_Q$ (selbe Triple-Orientierungen)

Typen von Punktmengen

$n = 4$:



abc	abd	acd	bcd
$+$	$+$	$+$	$+$

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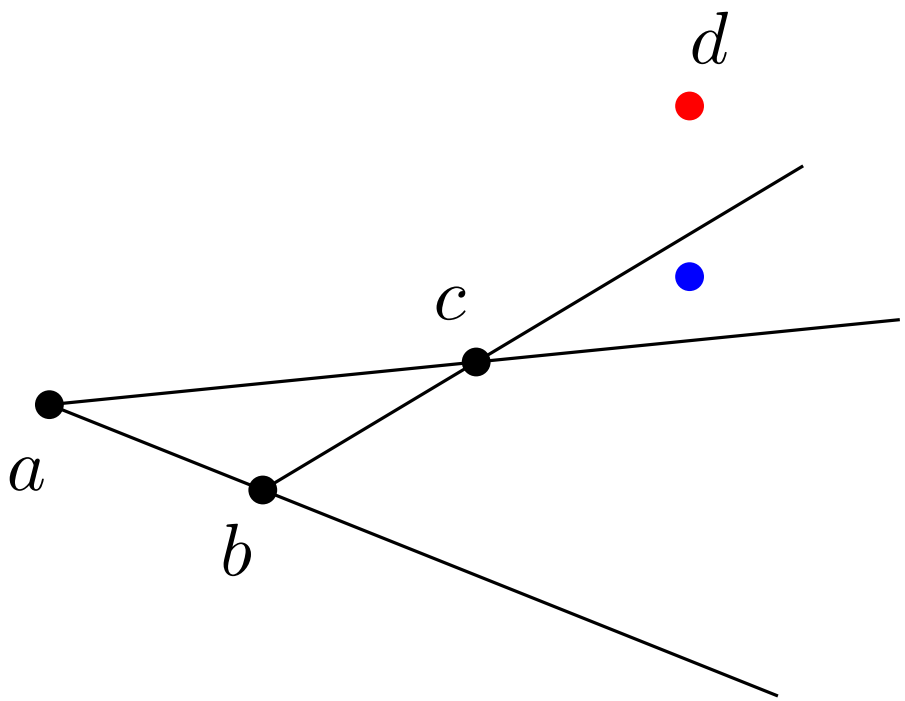
$n = 4$:



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$+$	$+$	$+$	$-$

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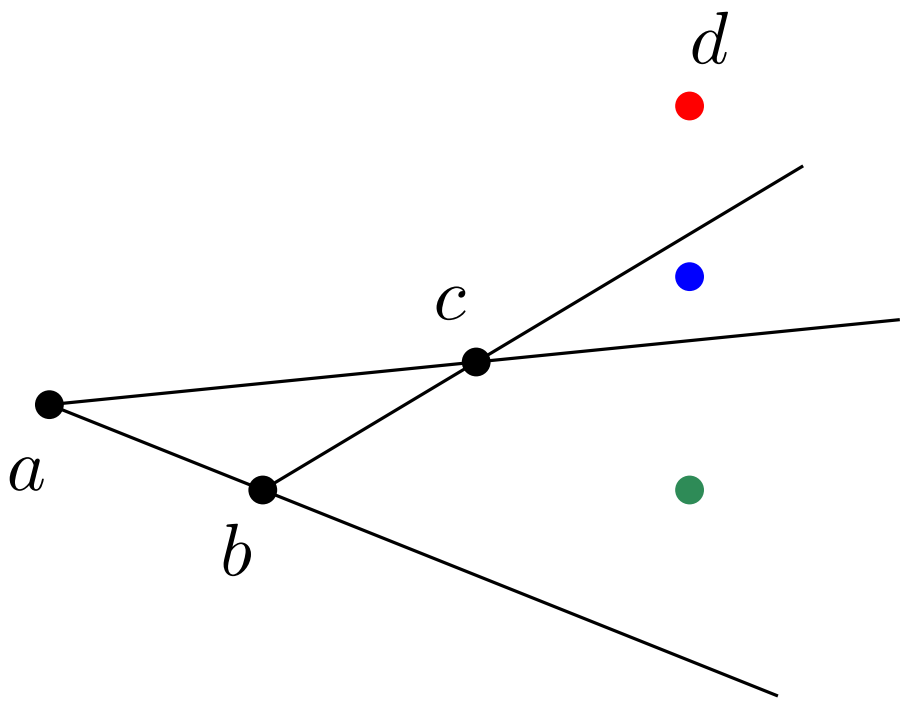

abc	abd	acd	bcd
$+$	$+$	$+$	$+$
$+$	$+$	$+$	$-$
$+$	$+$	$-$	$-$

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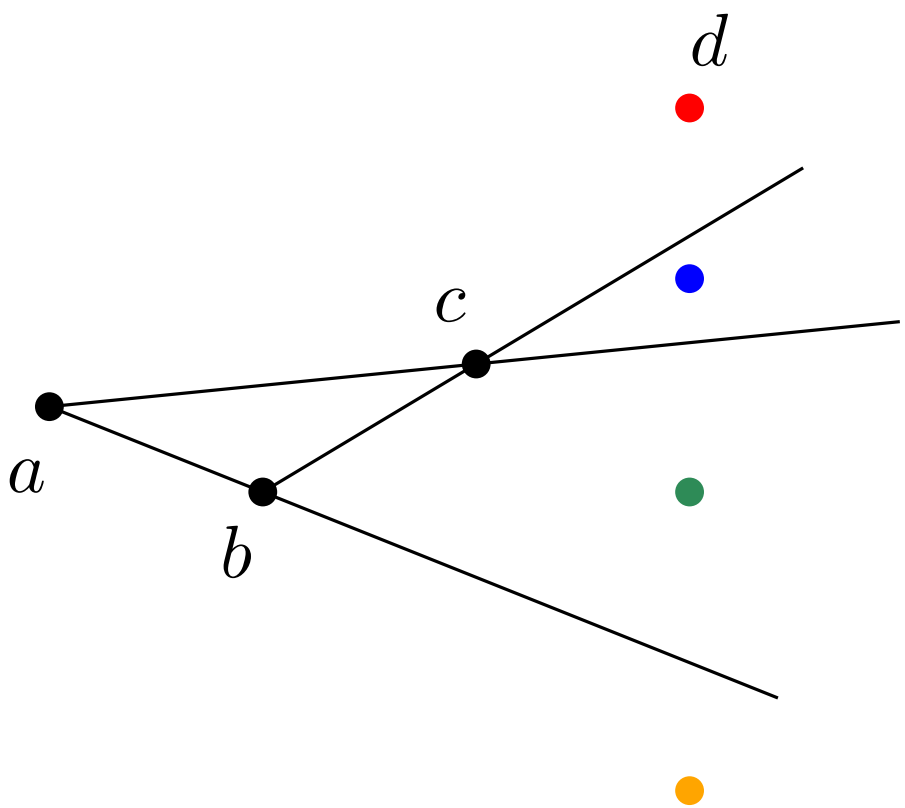
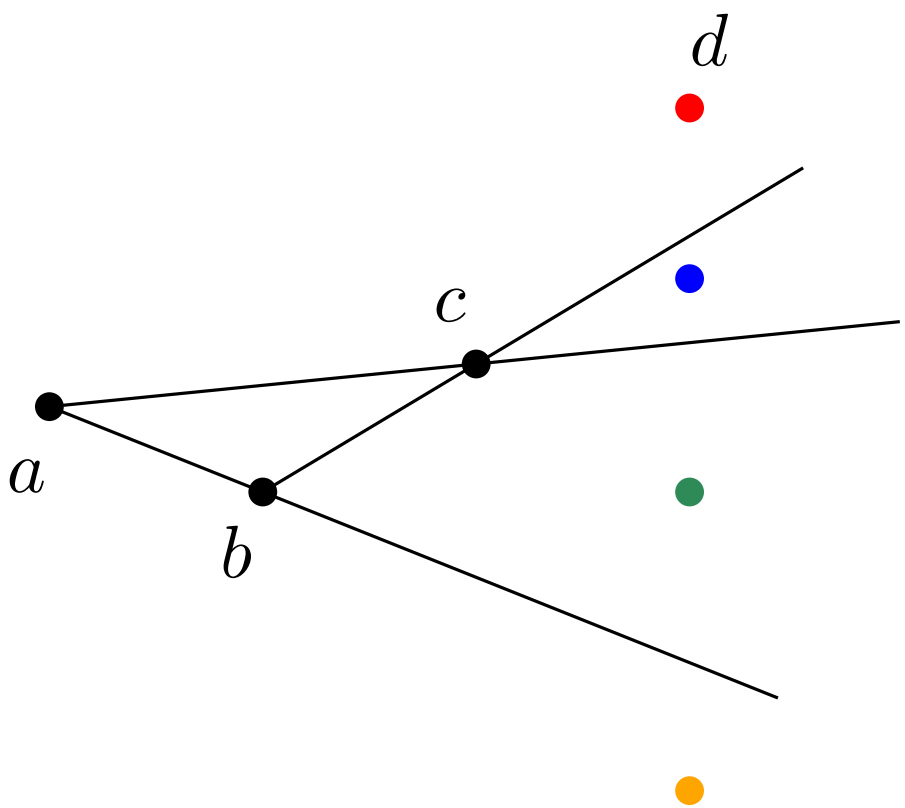


abc	abd	acd	bcd
$+$	$+$	$+$	$+$
$+$	$+$	$+$	$-$
$+$	$+$	$-$	$-$
$+$	$-$	$-$	$-$

Typen von Punktmengen

$n = 4$:



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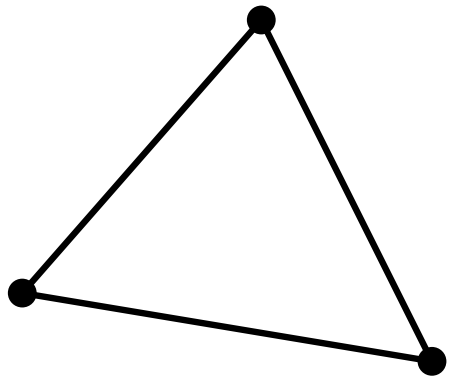
$+$	$+$	$+$	$-$
$+$	$+$	$-$	$-$
$+$	$-$	$-$	$-$
$-$	$-$	$-$	$-$
$-$	$-$	$-$	$+$
$-$	$-$	$+$	$+$
$-$	$+$	$+$	$+$

Typen von Punktmengen

$n = 3$:



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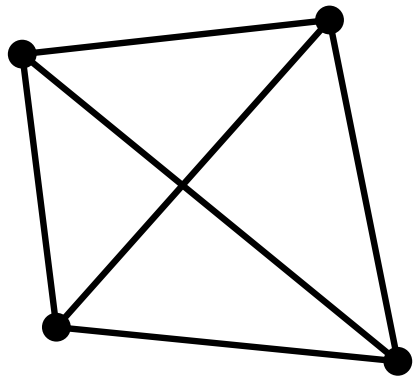

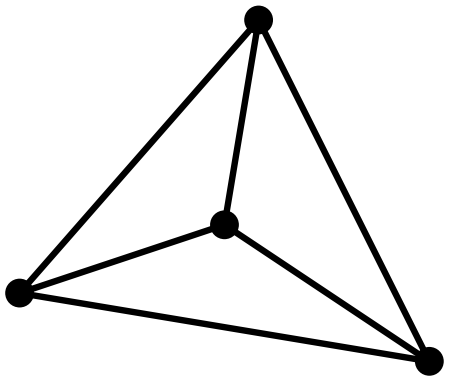

Typen von Punktmengen

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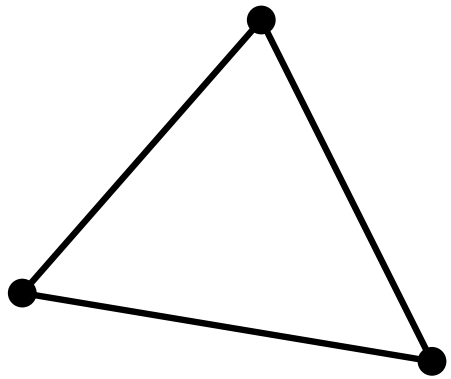

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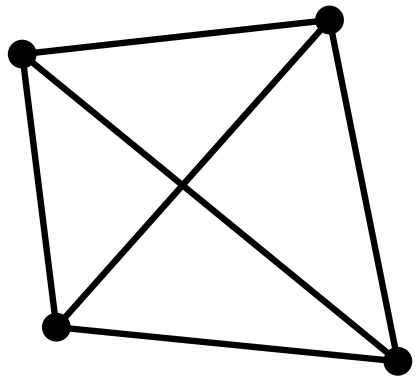

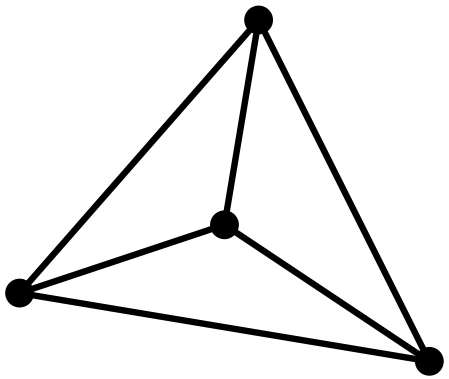

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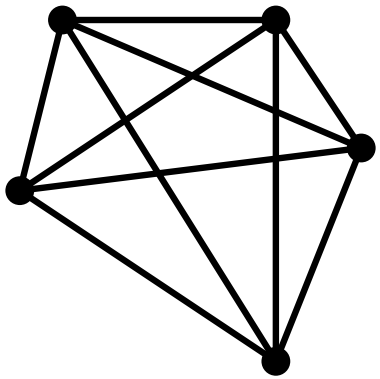
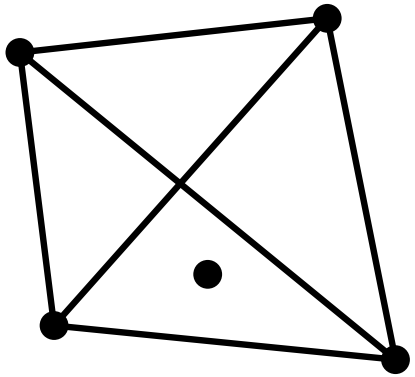
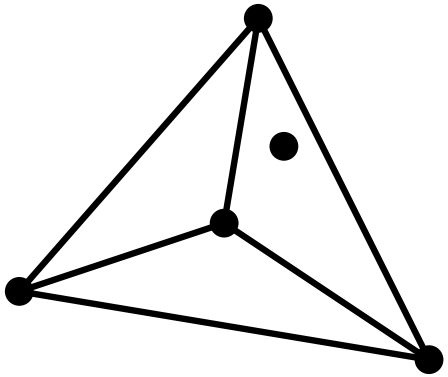
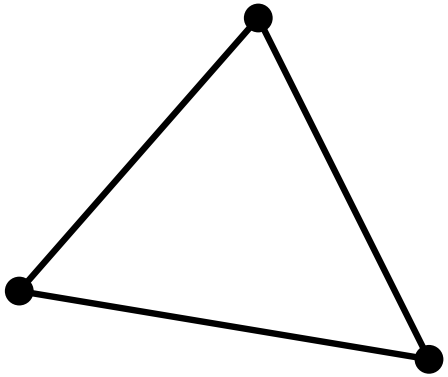






Typen von Punktmengen

$n = 3$:



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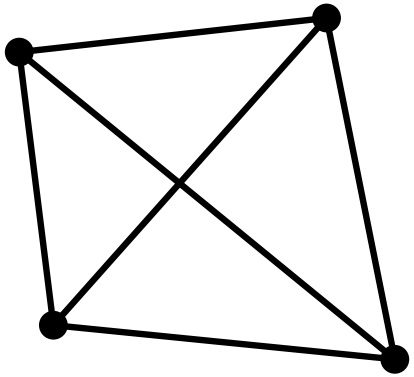
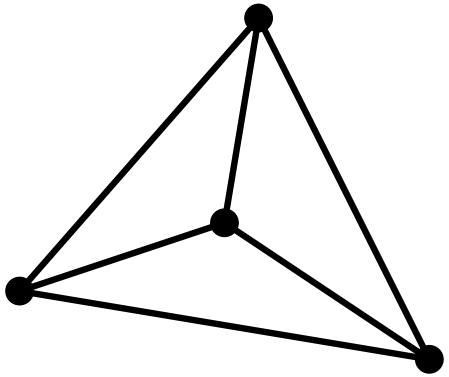
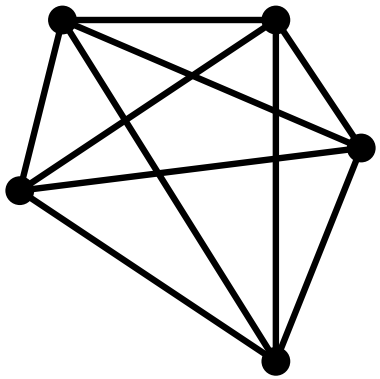
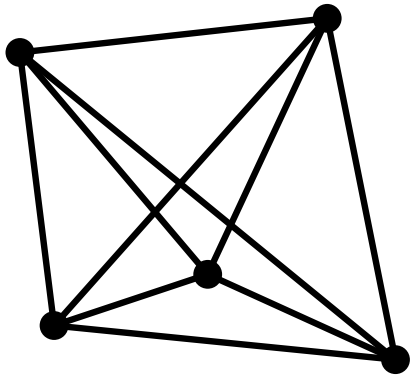
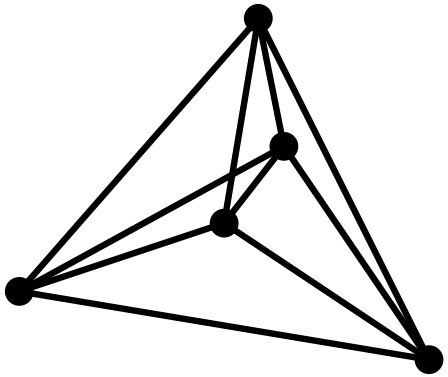




$n = 5$:







Typen von Punktmengen

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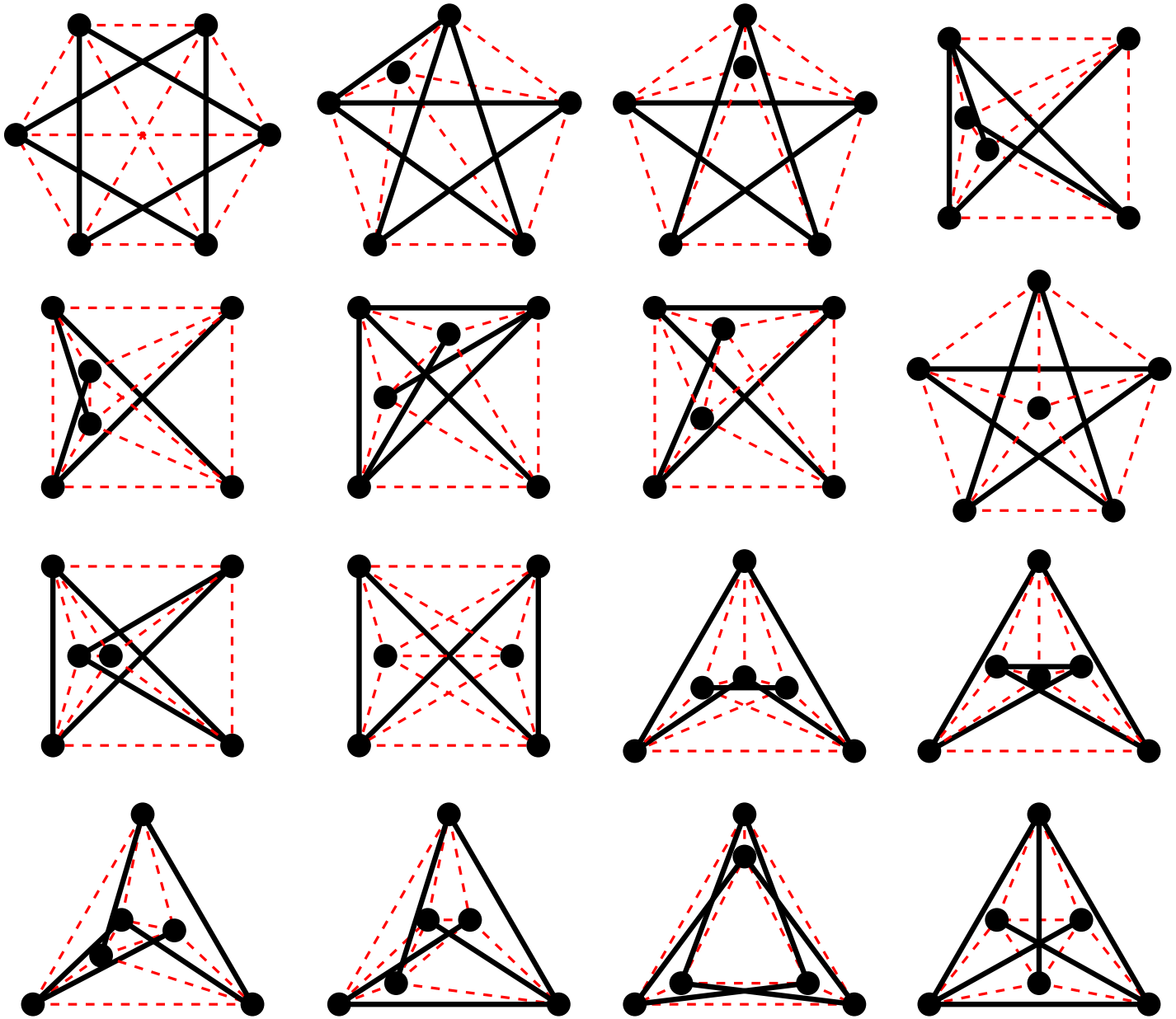


The OEIS is supported by the many generous donors to the OEIS Foundation.

THE ON-LINE ENCYCLOPEDIA OF INTEGER SEQUENCES ®

founded in 1964 by N. J. A. Sloane

Search Hints

(Greetings from The On-Line Encyclopedia of Integer Sequences!)

A063666 Euclidean order types: number of realizable order types of n points in the plane. 6

1, 2, 3, 16, 135, 3315, 158817, 14309547, 2334512907 (list; graph; refs; listen; history; text; internal format)

OFFSET 3,2

REFERENCES O. Aichholzer, F. Aurenhammer and H. Krasser. Enumerating order types for small point sets with applications. In Proc. 17th Ann. ACM Symp. Computational Geometry, pages 11-18, Medford, Massachusetts, USA, 2001.

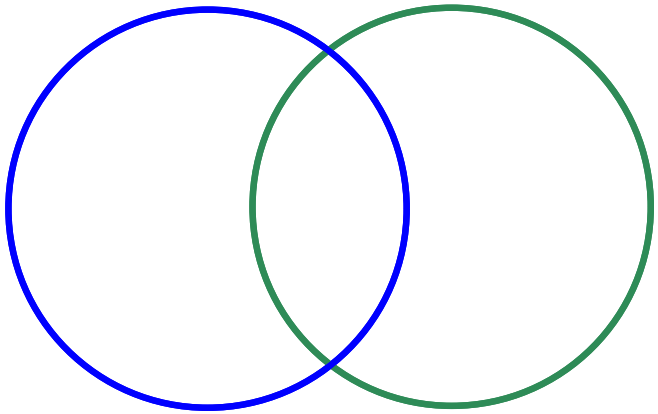
LINKS Table of n, a(n) for n=3..11.

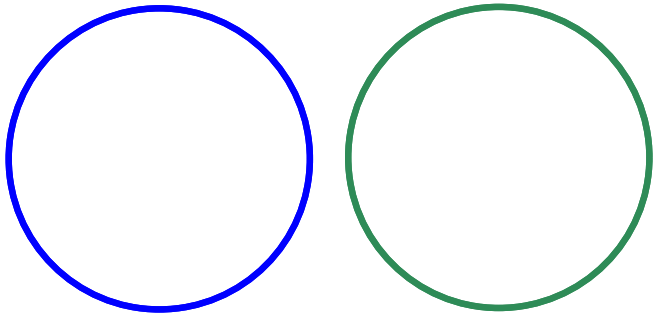
O. Aichholzer, F. Aurenhammer and H. Krasser, Enumerating order types for small point sets with applications

S. Felsner and J. E. Goodman, Pseudoline Arrangements. In: Toth, O'Rourke, Goodman (eds.) Handbook of Discrete and Computational Geometry, 3rd edn. CRC Press, 2018.

FORMULA Asymptotics: a(n) = 2^(Theta(n log n)). This is Bachmann-Landau notation, that is, there are constants n_0, c

Typen von Kreisarrangements





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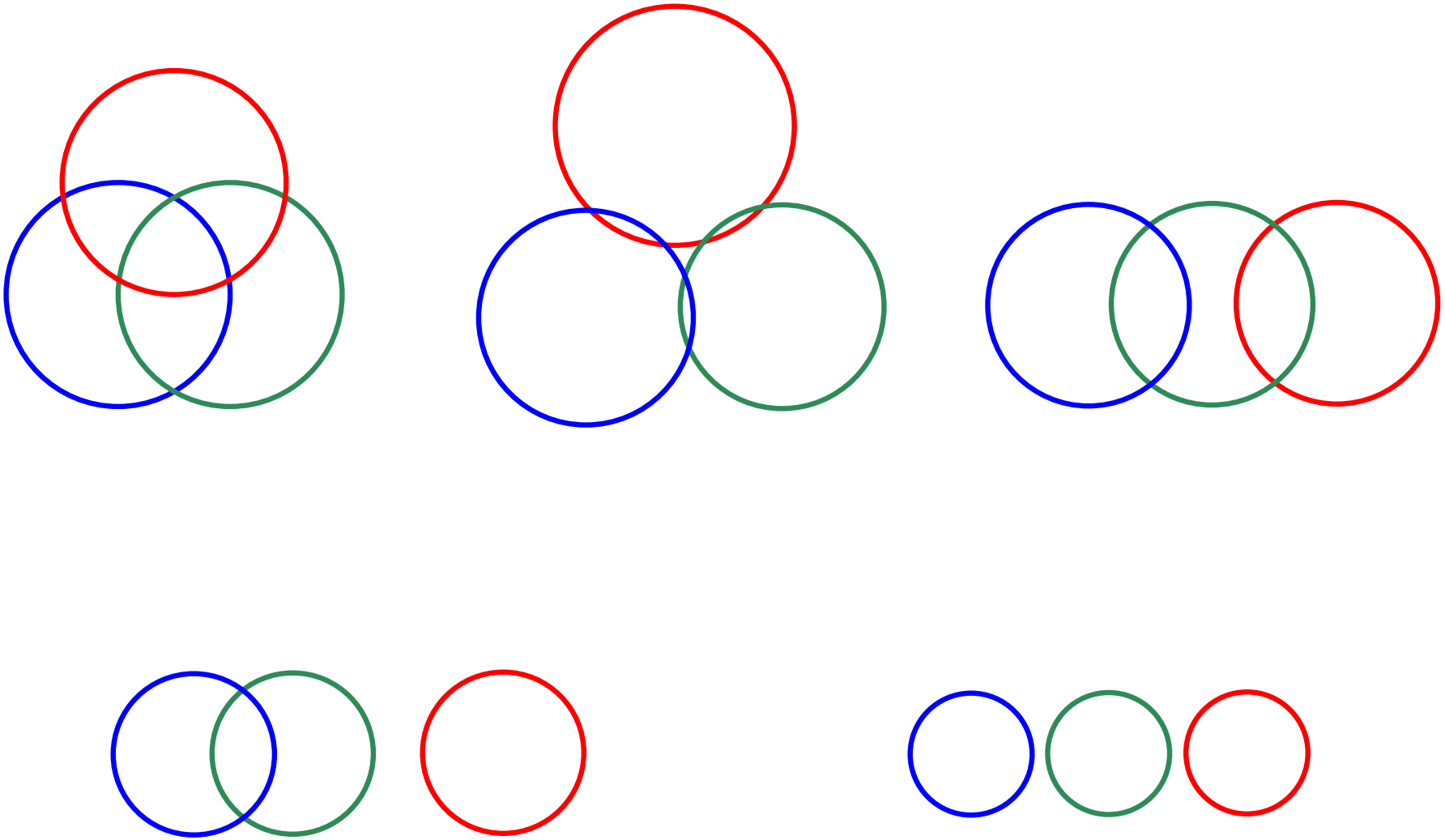

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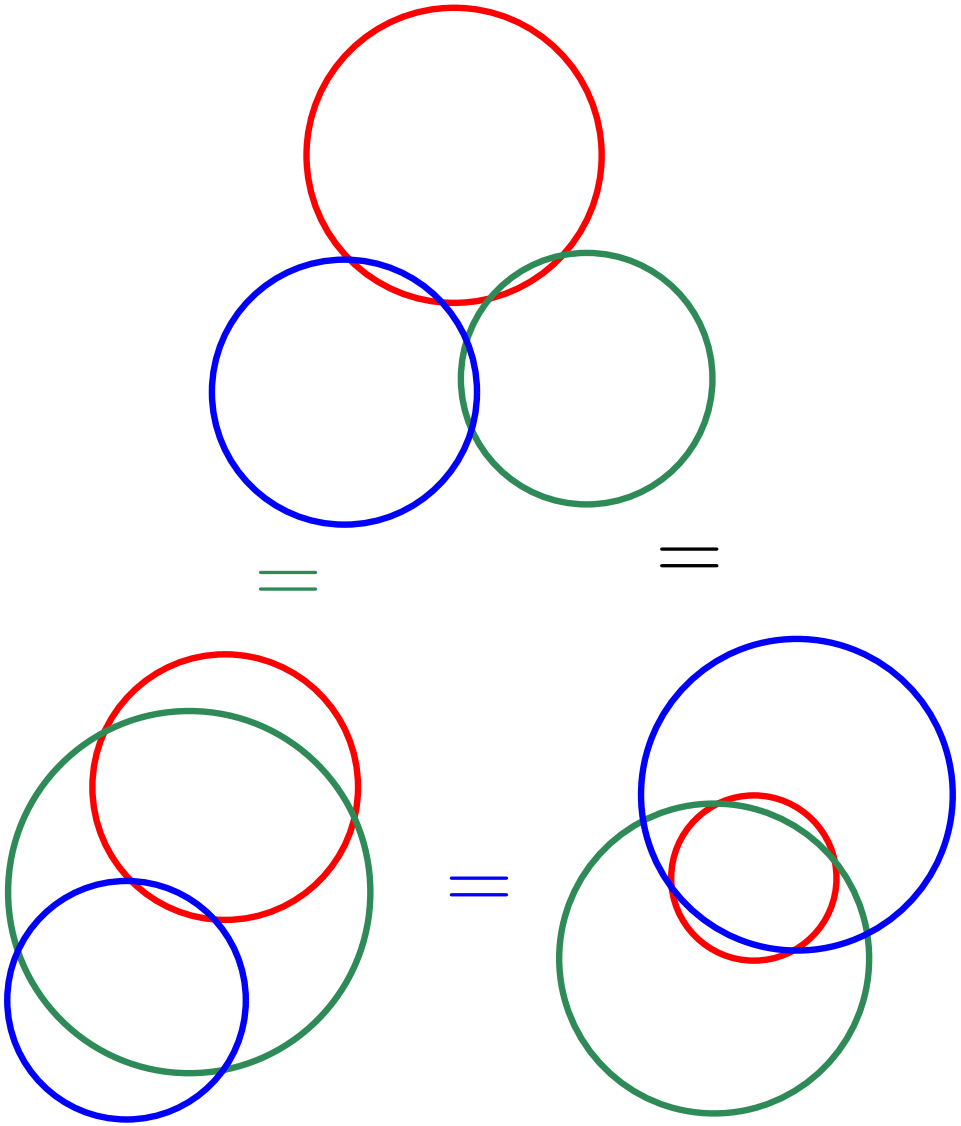

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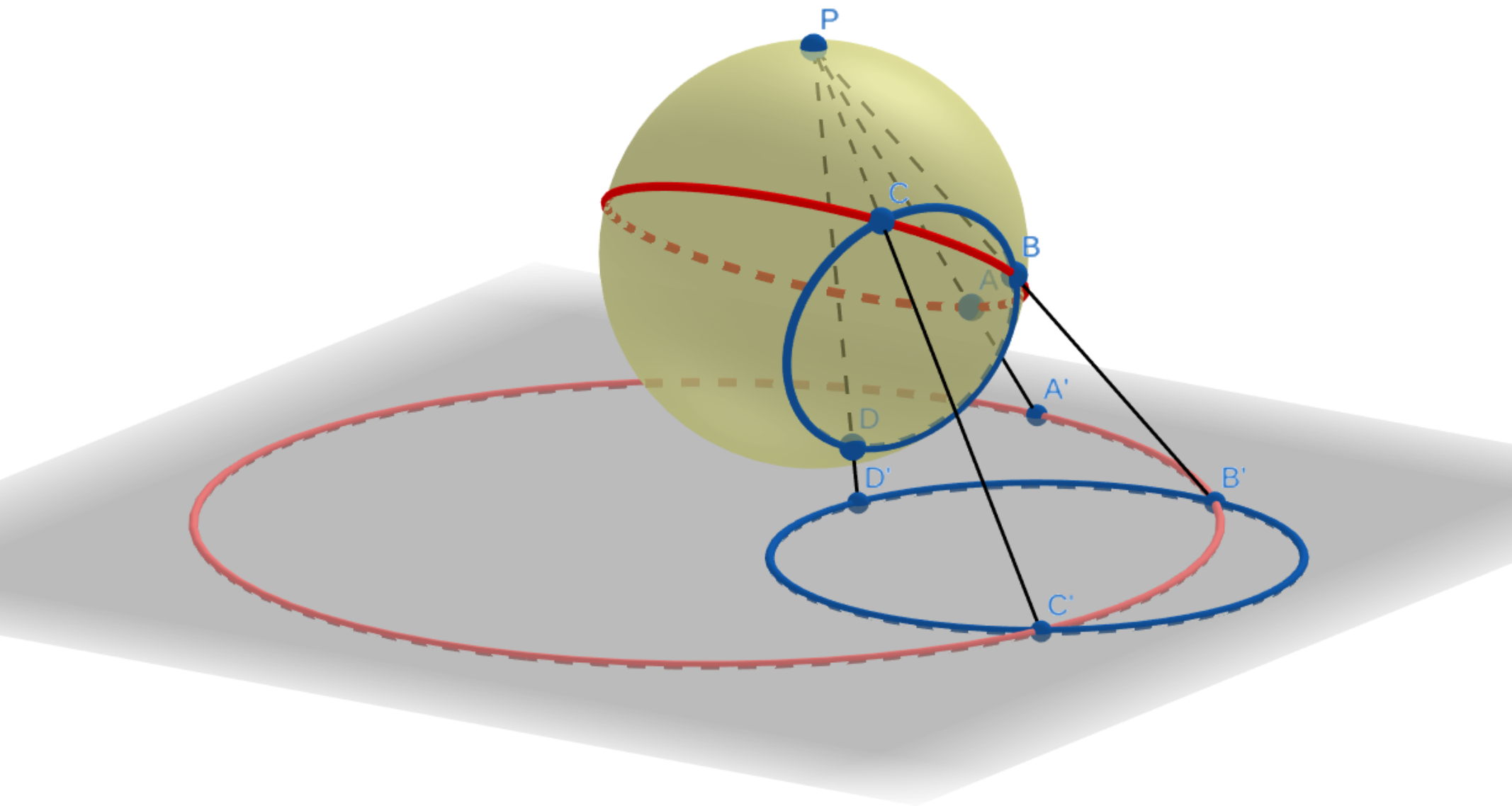

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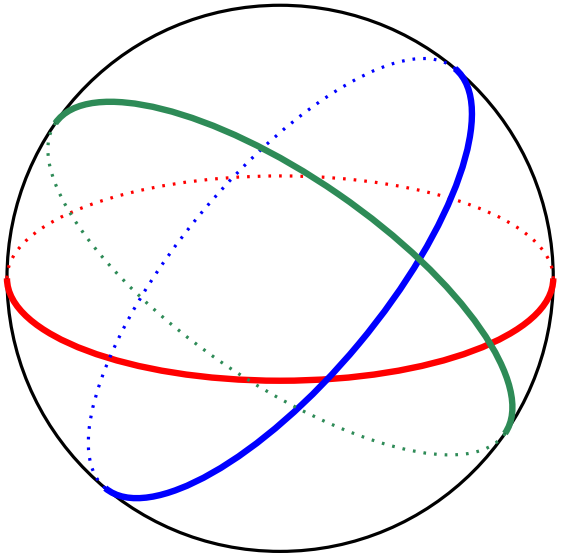
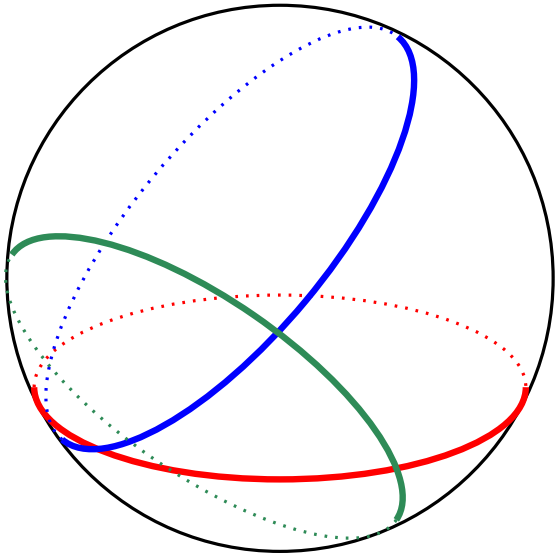
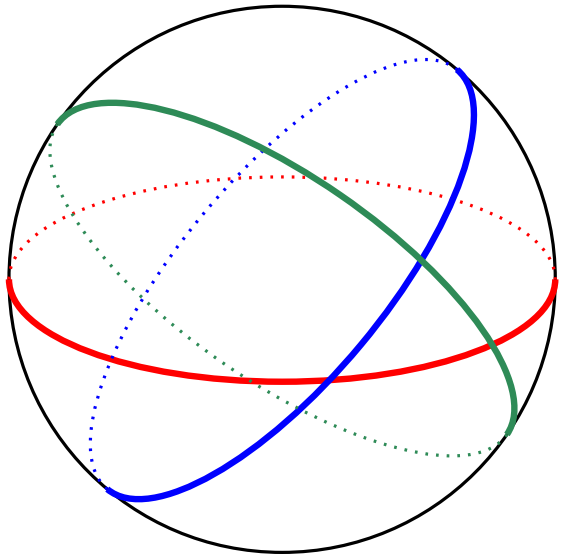


Großkreise



Kreise

Typen von Kreisarrangements



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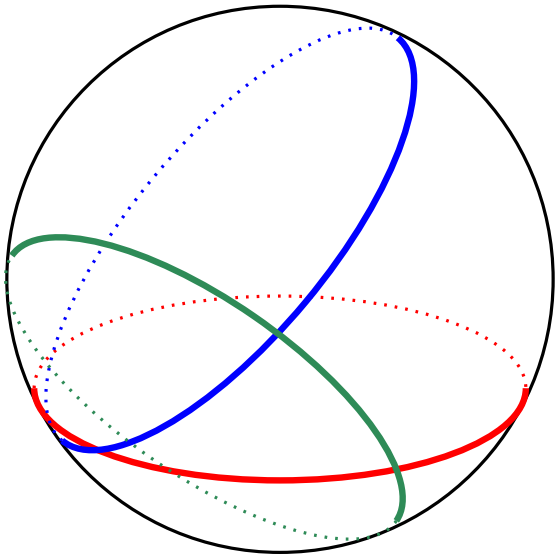

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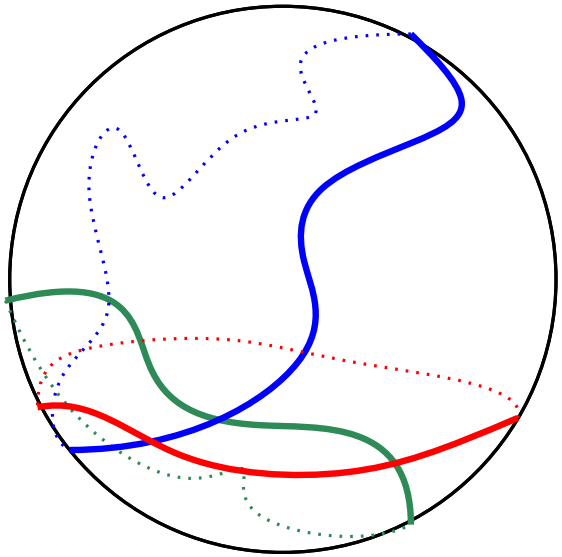
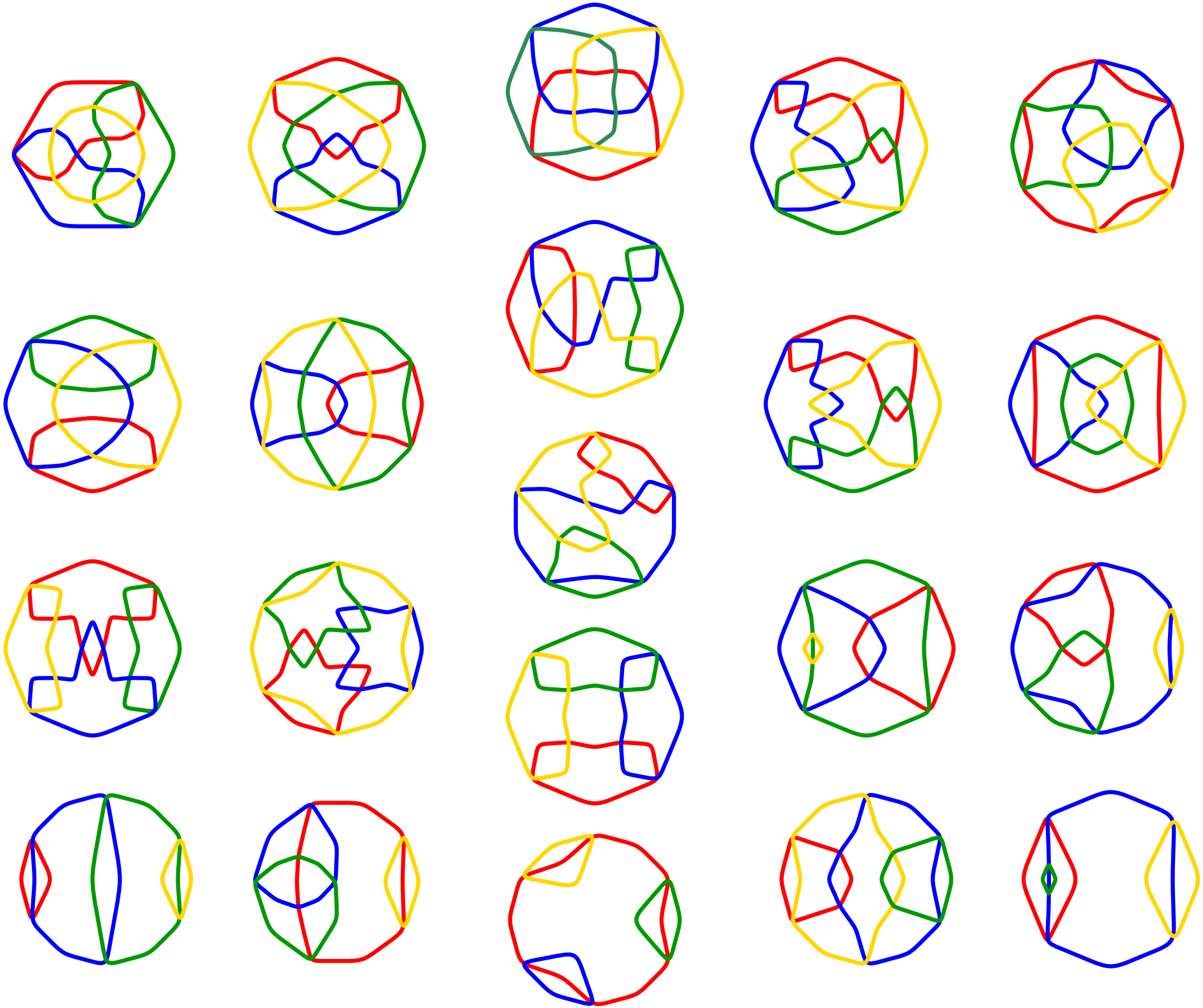


Pseudokreise



Vielen Dank für Eure Aufmerksamkeit!